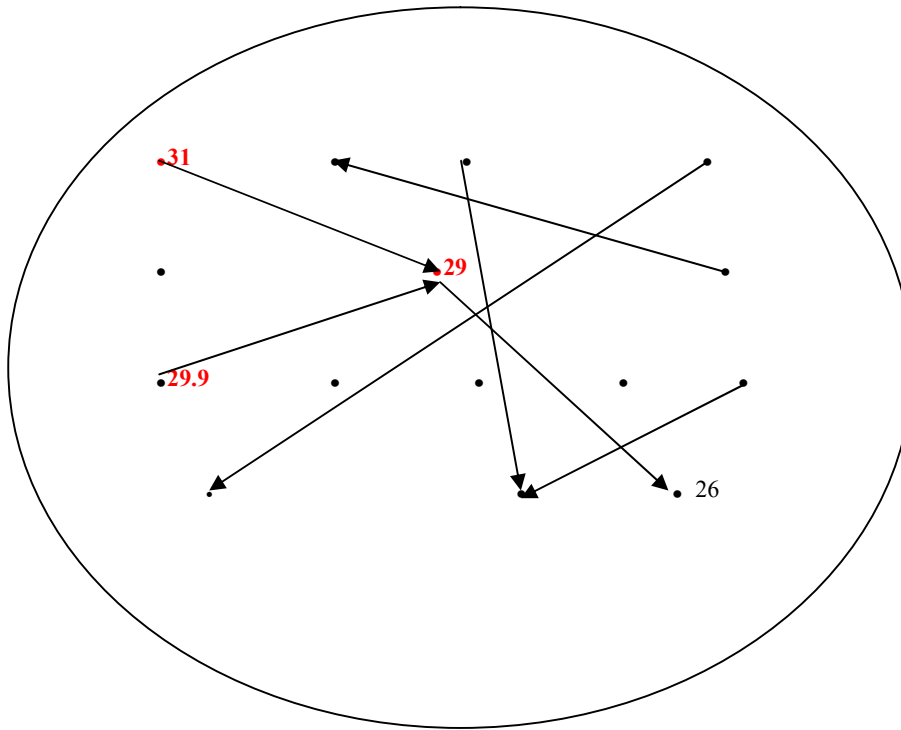


Gradient, Divergence & Curl

Gradient:

- Measures the rate of change in a scalar field; the gradient of a scalar field is a vector field. The derivative/differentiation/rate of change of a scalar field result in a vector field called the gradient.
- Computes the gradient of a scalar function. That is, it finds the Gradient, the slope, how fast you change, in any given direction.
- A gradient is applied to a scalar quantity that is a function of a 3D vector field: position. The gradient measures the direction in which the scalar quantity changes the most, as well as the rate of change with respect to position. A common example of this is height as a function of latitude and longitude, often applied to mountain ranges. A measure of the slope, and direction of the slope, is often called the gradient.



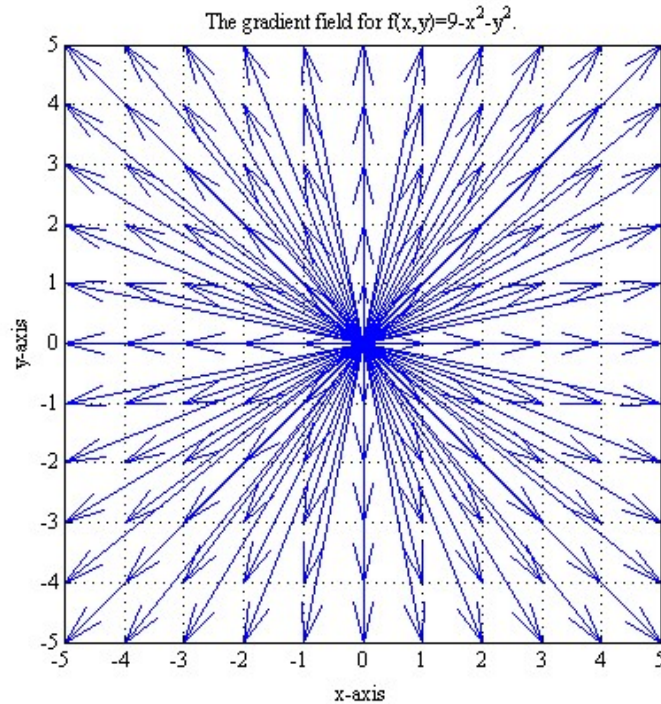


Figure # 58

Divergence:

- Measures a vector field's tendency to originate from or **convergent** upon a given point.
- Computes the divergence of a vector function. That is, it finds how much "stuff" is leaving a point in space.
- A divergence is applied to a vector as a function of position, yielding a scalar. The divergence actually measures how much the vector function is spreading out. If you are at a location from which the vector field tends to point away in all directions, you will definitely have a positive divergence. If the field points inward toward a point, the divergence in and near that point is negative. If just as much of the vector field points in as out, the divergence will be approximately zero.
- If we again think of $\vec{F}$ as the velocity field of a flowing fluid then $\text{div } \vec{F}$ represents the net rate of change of the mass of the fluid flowing from the point (x,y,z) per unit volume. This can also be thought of as the tendency of a fluid to diverge from a point.

The **divergence** of a vector field is relatively easy to understand intuitively. Imagine that the vector field $\vec{F}$ below gives the velocity of some fluid flow. It appears that the fluid is exploding outward from the origin

This expansion of fluid flowing with velocity field $\vec{F}$ is captured by the divergence of $\vec{F}$, which we denote $\text{div } \vec{F}$. The divergence of the above vector field is positive since the flow is expanding.

In contrast, the below vector field represents fluid flowing so that it compresses as it moves toward the origin. Since this compression of fluid is the opposite of expansion, the divergence of this vector field is negative.

A three-dimensional vector field $\vec{F}$ showing expansion of fluid flow is shown below.

Again, because of the expansion, we can conclude that $\text{div } \vec{F} > 0$.

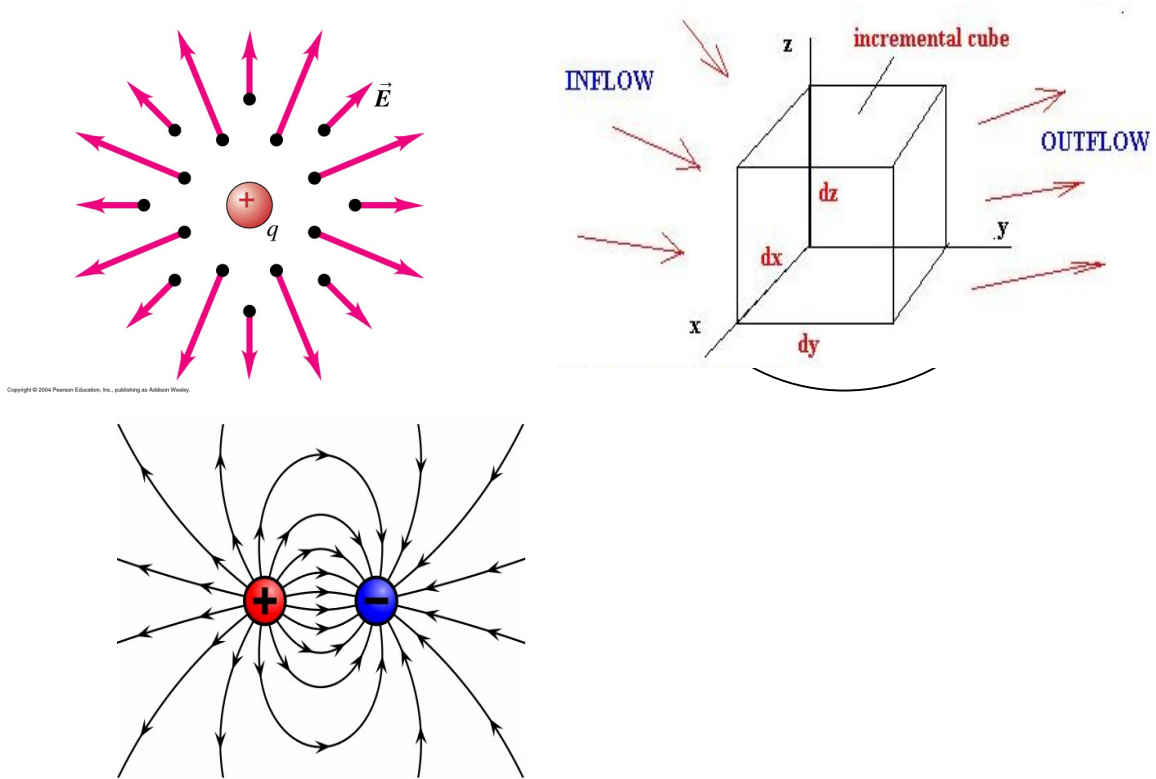


Figure # 59: Divergence of vectors flow field

[কোন একটি point এ চার্জের intensity/effect হচ্ছে divergence যেমন: কোন একটি point এ heat দিলাম। যেমন: কোন একটি point p এ heat দিলে তা চারিদিকে ছড়িয়ে পড়বে। q point এ তার intensity/effect কত? এটাই divergence]

Curl:

- In vector calculus, the **curl** (or **rotor**) is a vector operator that describes the infinitesimal rotation of a 3-dimensional vector field. At every point in the field, the curl is represented by a vector. The attributes of this vector (length and direction) characterize the rotation at that point.
- The direction of the curl is the axis of rotation, as determined by the right-hand rule, and the magnitude of the curl is the magnitude of rotation. If the vector field represents the flow velocity of a moving fluid, then the curl is the **circulation density** of the fluid. A vector field whose curl is zero is called irrotational. The curl is a form of differentiation for vector fields

- measures a vector field's tendency to rotate about a point; the curl of a vector field is another vector field.
- It computes the rotational aspects of a vector function, maybe people thought how vectors "curl" around a center point, like wind curling around a low pressure on a weather map.
- A curl measures just that, the curl of a vector field. Unlike the divergence, a curl yields a vector. A vector field that tends to point around an axis, such as vectors pointing tangential to a circle, will yield a non-zero curl with the axis around which the curl occurs as the direction. Another example is the velocity field of motion spiraling in or out, such as a whirlpool. Point your right-hand thumb along the direction of the curl. Curl your fingers around this axis. They will curl in the same direction as the vector field. I do not know the names of the texts, but I know there are books available with vector fields to illustrate both divergence and curl.

[একটি field এ কি পরিমান twist/wrapping (পাঁচ) আছে তা measurement করাই curling]

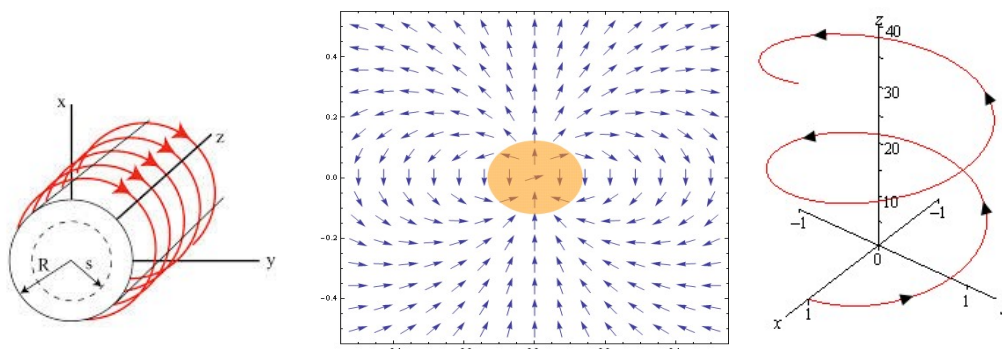


Figure # 60

Mathematical Expression of Gradient, divergence, curl of a Vector Field

Vector differential operator : $\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$ and is denoted by the symbol $\vec{\nabla}$ (pronounced 'del' or sometimes 'Nabla')

That is $\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$ -----(i)

Beware! ∇ Cannot exist alone: it is an operator and must operate on a stated scalar function $\phi(x, y, z)$.

If F is a vector, ∇F has no meaning

Grad (gradient of a scalar function)

If a scalar function $\phi(x, y, z)$ is continuously differentiable with respect to its variables x, y, z , throughout the region, then the *gradient* of ϕ , Written $\text{grad } \phi$, is defined as the vector

$$\text{grad } \phi = \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \text{ -----(ii) where } \phi \text{ is a function}$$

of x, y, z

Note that, while ϕ is a scalar function, $\text{grad } \phi$ is a vector function

Divergence of a vector field: If we form the scalar (dot) product of $\vec{\nabla}$ with a vector function $\vec{A}(x, y, z) = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ we get a scalar result called the divergence of $\vec{A}$:

$$\text{div } \vec{A} \equiv \vec{\nabla} \cdot \vec{A} \equiv \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (A_x \hat{i} + A_y \hat{j} + A_z \hat{k})$$

$$\text{div } \vec{A} \equiv \vec{\nabla} \cdot \vec{A} \equiv \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \text{ -----(iii)}$$

Curl of a vector field: The curl of a vector field $\vec{A}(x, y, z) = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ is defined by

$$\vec{\nabla} \times \vec{A} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \text{ -----(iv)}$$

Q# 23: If $\phi(x, y, z) = 3x^2y - y^3z^2$, find $\vec{\nabla} \Phi$ (or $\text{grad } \Phi$) at the point $(1, -2, -1)$.

$$\text{Answer: } \vec{\nabla} \Phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (3x^2y - y^3z^2)$$

$$= \hat{i} \frac{\partial}{\partial x} (3x^2y - y^3z^2) + \hat{j} \frac{\partial}{\partial y} (3x^2y - y^3z^2) + \hat{k} \frac{\partial}{\partial z} (3x^2y - y^3z^2)$$

$$= \hat{i} (6xy - 0) + \hat{j} (3x^2 - 3y^2z^2) + \hat{k} (0 - 2y^3z)$$

$$= 6xy \hat{i} + (3x^2 - 3y^2z^2) \hat{j} - 2y^3z \hat{k}$$

$$= 6(1)(-2) \hat{i} + \{3(1)^2 - 3(-2)^2(-1)^2\} \hat{j} - 2(-2)^3(-1) \hat{k}$$

$$= -12 \hat{i} - 9 \hat{j} - 16 \hat{k} \text{ (Answer).}$$

Q# 24: Find $\vec{\nabla} \phi$ if (a) $\phi = \ln |\vec{r}|$ (b) $\phi = \frac{1}{|\vec{r}|}$

$$(a) \text{ Let } \vec{r} = x \hat{i} + y \hat{j} + z \hat{k} \text{ Then } |\vec{r}| = \sqrt{x^2 + y^2 + z^2} \text{ and } |\vec{r}|^2 = x^2 + y^2 + z^2$$

$$\begin{aligned}
\text{Then } \phi &= \ln \left| \vec{r} \right| = \ln \sqrt{x^2 + y^2 + z^2} = \ln (x^2 + y^2 + z^2)^{1/2} = \frac{1}{2} \ln(x^2 + y^2 + z^2) \\
\therefore \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial}{\partial x} \phi + \hat{j} \frac{\partial}{\partial y} \phi + \hat{k} \frac{\partial}{\partial z} \phi \\
&= \frac{1}{2} \hat{i} \frac{\partial}{\partial x} \ln(x^2 + y^2 + z^2) + \frac{1}{2} \hat{j} \frac{\partial}{\partial y} \ln(x^2 + y^2 + z^2) + \frac{1}{2} \hat{k} \frac{\partial}{\partial z} \ln(x^2 + y^2 + z^2) \\
&= \frac{1}{2} \hat{i} \left(\frac{1}{x^2 + y^2 + z^2} \right) \left\{ \frac{\partial}{\partial x} (x^2 + y^2 + z^2) \right\} + \frac{1}{2} \hat{j} \left(\frac{1}{x^2 + y^2 + z^2} \right) \left\{ \frac{\partial}{\partial y} (x^2 + y^2 + z^2) \right\} + \frac{1}{2} \\
&\quad \hat{k} \left(\frac{1}{x^2 + y^2 + z^2} \right) \left\{ \frac{\partial}{\partial z} (x^2 + y^2 + z^2) \right\} \\
&= \frac{1}{2} \hat{i} \left(\frac{1}{x^2 + y^2 + z^2} \right) (2x + 0 + 0) + \frac{1}{2} \hat{j} \left(\frac{1}{x^2 + y^2 + z^2} \right) (0 + 2y + 0) + \frac{1}{2} \\
&\quad \hat{k} \left(\frac{1}{x^2 + y^2 + z^2} \right) (0 + 0 + 2z) \\
&= \frac{1}{2} \hat{i} \left(\frac{2x}{x^2 + y^2 + z^2} \right) + \frac{1}{2} \hat{j} \left(\frac{2y}{x^2 + y^2 + z^2} \right) + \frac{1}{2} \hat{k} \left(\frac{2z}{x^2 + y^2 + z^2} \right) \\
&= \frac{1}{2} \times 2 \left\{ \frac{x \hat{i} + y \hat{j} + z \hat{k}}{x^2 + y^2 + z^2} \right\} = \left\{ \frac{x \hat{i} + y \hat{j} + z \hat{k}}{x^2 + y^2 + z^2} \right\} = \frac{\vec{r}}{\left| \vec{r} \right|^2} \text{ Answer}
\end{aligned}$$

$$(b) \text{ Given, } \phi = \frac{1}{\left| \vec{r} \right|} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} = (x^2 + y^2 + z^2)^{-1/2}$$

$$\begin{aligned}
\vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial}{\partial x} \phi + \hat{j} \frac{\partial}{\partial y} \phi + \hat{k} \frac{\partial}{\partial z} \phi \\
&= \hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2)^{-1/2} + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2)^{-1/2} \\
&= \hat{i} \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-1/2-1} \frac{\partial}{\partial x} (x^2 + y^2 + z^2) + \hat{j} \left(-\frac{1}{2} \right) \\
&\quad (x^2 + y^2 + z^2)^{-1/2-1} \frac{\partial}{\partial y} (x^2 + y^2 + z^2) + \hat{k} \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-1/2-1} \\
&\quad \frac{\partial}{\partial z} (x^2 + y^2 + z^2) \\
&= \hat{i} \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-1/2-1} (2x + 0 + 0) + \hat{j} \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-1/2-1} \\
&\quad (0 + 2y + 0) + \hat{k} \left(-\frac{1}{2} \right) (x^2 + y^2 + z^2)^{-1/2-1} \frac{\partial}{\partial z} (0 + 0 + 2z)
\end{aligned}$$

$$\begin{aligned}
&= \hat{i}(-x)(x^2+y^2+z^2)^{-3/2} + \hat{j}(-y)(x^2+y^2+z^2)^{-3/2} + \hat{k}(-z)(x^2+y^2+z^2)^{-3/2} \\
&= \frac{-x\hat{i}}{(x^2+y^2+z^2)^{3/2}} + \frac{-y\hat{j}}{(x^2+y^2+z^2)^{3/2}} + \frac{-z\hat{k}}{(x^2+y^2+z^2)^{3/2}} \\
&= -\frac{(x\hat{i}+y\hat{j}+z\hat{k})}{(x^2+y^2+z^2)^{3/2}} = -\frac{(x\hat{i}+y\hat{j}+z\hat{k})}{(x^2+y^2+z^2)^1 \cdot (x^2+y^2+z^2)^{1/2}} \\
&= -\frac{\vec{r}}{|\vec{r}|^2 |\vec{r}|} = -\frac{\vec{r}}{|\vec{r}|^3} \text{ Answer}
\end{aligned}$$

Q#25: Find the level curve of $f(x, y) = -x^2 + y^2$ passing through (2, 3). Draw Graph the gradient at the point (2, 3)

Answer: Given, $f(x, y) = -x^2 + y^2$

$$f(2, 3) = -2^2 + 3^2 = -4 + 9 = 5$$

Hence the level curve is the hyperbola, i.e.

$$f(x, y) = -x^2 + y^2 = 5$$

$$\text{i.e. } -x^2 + y^2 = 5$$

$$\text{i.e. } x^2 - y^2 = -5$$

$$\Rightarrow \frac{x^2}{-5} - \frac{y^2}{-5} = 1 \text{ [This is the equation of a hyperbola, i. e. } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1] \text{-----(i)}$$

From (i),

$$\Rightarrow \frac{x^2}{-5} - \frac{y^2}{-5} = 1 \quad \Rightarrow x^2 - y^2 = -5 \quad \Rightarrow y^2 = 5 + x^2 \quad \Rightarrow y = \pm\sqrt{5+x^2}$$

$$\Rightarrow y = \pm\sqrt{5+x^2} \text{-----(ii)}$$

x	0	-1	-2	-3	1	2	3	-4	4	
$y = \pm\sqrt{5+x^2}$	$\pm\sqrt{5}$ $= \pm 2.23$	$\pm\sqrt{6}$ ± 2.44	± 3 ± 3	$\pm\sqrt{14}$ ± 3.74	$\pm\sqrt{6}$ ± 2.44	± 3 ± 3	$\pm\sqrt{14}$ ± 3.74	$\pm\sqrt{19}$ ± 4.35	$\pm\sqrt{19}$ ± 4.35	

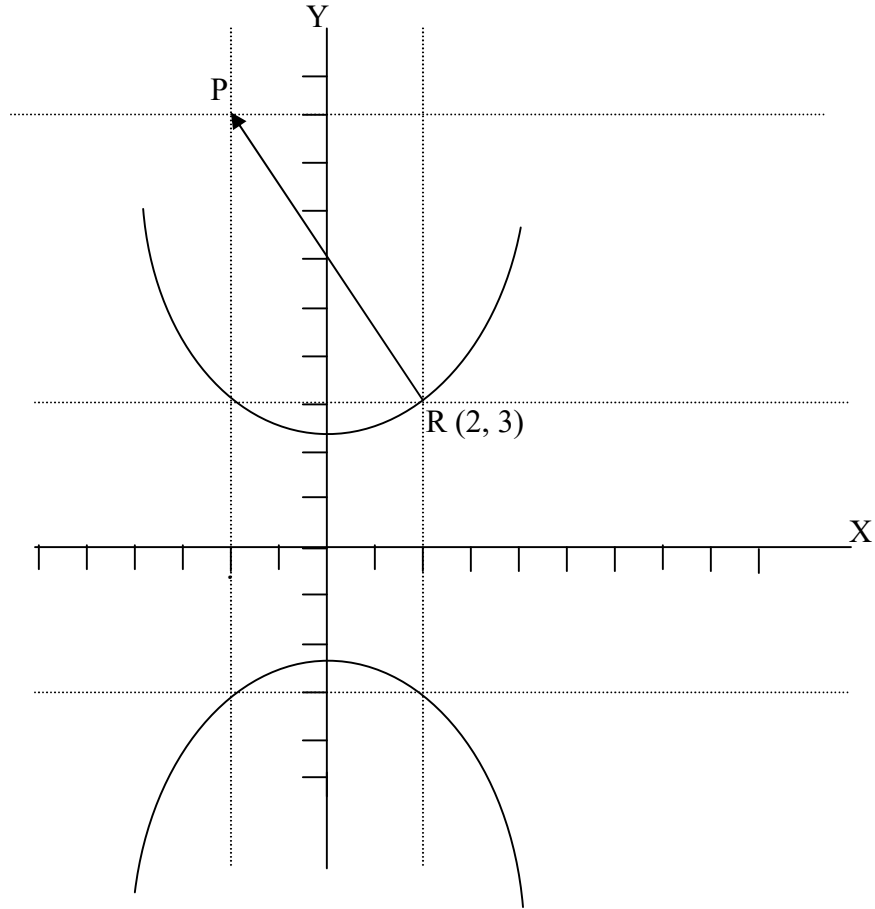


Figure # 61

Given, $f(x, y) = -x^2 + y^2$

Now, Gradient of the function, i.e.

$$\vec{\nabla} f(x, y) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (-x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (-x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (-x^2 + y^2) + \hat{j} \frac{\partial}{\partial y} (-x^2 + y^2) + \hat{k} \frac{\partial}{\partial z} (-x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (-x^2) + \hat{i} \frac{\partial}{\partial x} (y^2) + \hat{j} \frac{\partial}{\partial y} (-x^2) + \hat{j} \frac{\partial}{\partial y} (y^2) + \hat{k} \frac{\partial}{\partial z} (-x^2) + \hat{k} \frac{\partial}{\partial z} (y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i}(-2x) + \hat{i} \times 0 + \hat{j} \times 0 + \hat{j}(2y) + \hat{k} \times 0 + \hat{k} \times 0$$

$$\vec{\nabla} f(x, y) = -2x \hat{i} + 2y \hat{j}$$

$$\vec{\nabla} f(2, 3) = -2 \times 2 \hat{i} + 2 \times 3 \hat{j}$$

$$\vec{\nabla} f(2, 3) = -4 \hat{i} + 6 \hat{j}$$

Hence the gradient Vector is $\vec{RP} = \vec{\nabla} f(2, 3) = -4 \hat{i} + 6 \hat{j}$ the Answer

Q# 26: Sketch the level curve for the function $f(x, y) = x^2 + y^2$ through the point (3, 4) and draw the gradient vector at this point.

Answer: Given, the function $f(x, y) = x^2 + y^2$ through the point (3, 4),

$$f(3, 4) = 3^2 + 4^2$$

$$f(3, 4) = 9 + 16 = 25$$

Since $f(3, 4) = 25$, the level curve through the point (3, 4) has the equation

$f(x, y) = x^2 + y^2 = 25$, which is the circle. That is $x^2 + y^2 = 25$ whose centre (0, 0) and radius 5.

Now,

$$f(x, y) = x^2 + y^2$$

Now, Gradient of the function,

$$\vec{\nabla} f(x, y) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (x^2 + y^2) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i} \frac{\partial}{\partial x} (x^2) + \hat{i} \frac{\partial}{\partial x} (y^2) + \hat{j} \frac{\partial}{\partial y} (x^2) + \hat{j} \frac{\partial}{\partial y} (y^2) + \hat{k} \frac{\partial}{\partial z} (x^2) + \hat{k} \frac{\partial}{\partial z} (y^2)$$

$$\vec{\nabla} f(x, y) = \hat{i}(2x) + \hat{i} \times 0 + \hat{j} \times 0 + \hat{j}(2y) + \hat{k} \times 0 + \hat{k} \times 0$$

$$\vec{\nabla} f(x, y) = 2x \hat{i} + 2y \hat{j} \quad \text{------(i)}$$

The gradient vector at (3, 4) is

$$\therefore \vec{\nabla} f(3, 4) = 2 \times 3 \hat{i} + 2 \times 4 \hat{j}$$

$$\vec{\nabla} f(3, 4) = 6 \hat{i} + 8 \hat{j} \quad \text{------(ii)}$$

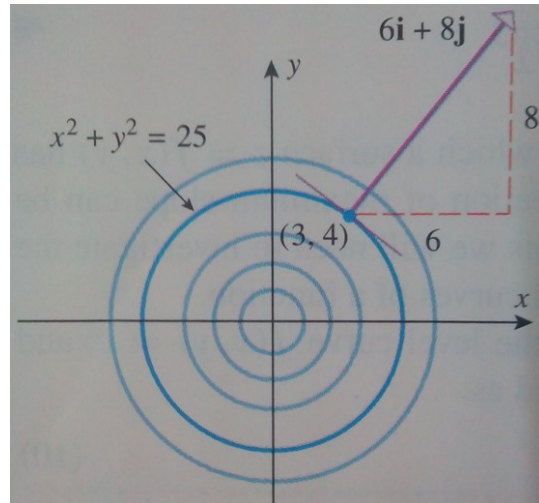


Figure # 62

Hence the gradient vector is perpendicular to the circle at $(3,4)$.

Q# 27: Sketch the gradient field of $\phi(x, y) = x + y$

Answer: Now, the gradient of the function $\phi(x, y) = x + y$

$$\vec{\nabla} \phi(x, y) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x + y)$$

$$\vec{\nabla} \phi(x, y) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x + y)$$

$$\vec{\nabla} \phi(x, y) = \hat{i} \frac{\partial}{\partial x} (x + y) + \hat{j} \frac{\partial}{\partial y} (x + y) + \hat{k} \frac{\partial}{\partial z} (x + y)$$

$$\vec{\nabla} \phi(x, y) = \hat{i}(1 + 0) + \hat{j}(0 + 1) + \hat{k}(0 + 0)$$

$$\vec{\nabla} \phi(x, y) = \hat{i} + \hat{j}$$

This is the same at each point. A portion of the vector field is sketched in figure below:

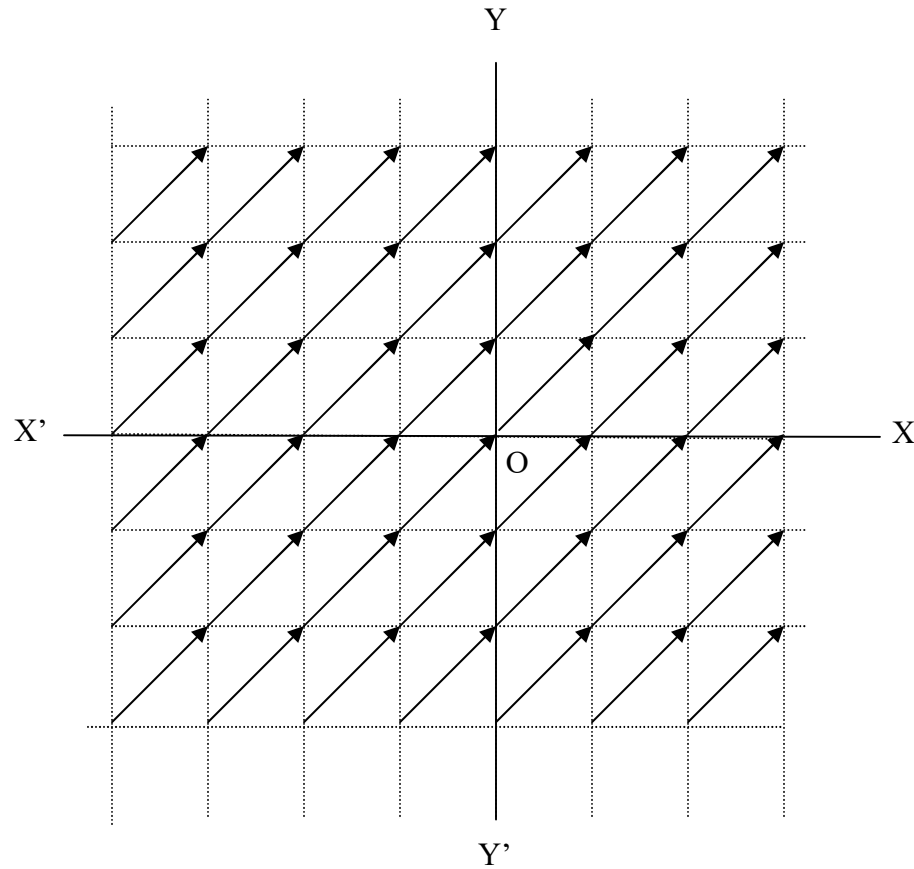


Figure: 63

Q# 28:

Given,

$$\phi(x, y) = x^2 y$$

$$\frac{\partial \phi}{\partial x} = 2xy \quad \text{-----(i)}$$

$$\frac{\partial \phi}{\partial y} = x^2 \times 1 \quad \text{-----(ii)}$$

We have,

$$\phi(x, y) = x^2 y$$

$$\therefore \frac{d\phi}{dx} = \frac{d}{dx}(x^2 y)$$

$$= x^2 \frac{d}{dx}(y) + y \frac{d}{dx}(x^2) \left[\because \frac{d}{dx}(uv) = u \frac{d}{dx}(v) + v \frac{d}{dx}(u) \right]$$

$$= x^2 \frac{dy}{dx} + y \times 2x \times 1$$

$$\therefore \frac{d\phi}{dx} = x^2 \frac{dy}{dx} + 2xy$$

$$\therefore \frac{d\phi}{dx} = 2xy + x^2 \frac{dy}{dx}$$

Again

$$\phi(x, y) = x^2 y$$

$$\therefore \frac{d\phi}{dy} = \frac{d}{dy}(x^2 y)$$

$$= x^2 \frac{d}{dy}(y) + y \frac{d}{dy}(x^2) [\therefore \frac{d}{dx}(uv) = u \frac{d}{dx}(v) + v \frac{d}{dx}(u)]$$

$$= x^2 \times 1 + y \times 2x \frac{dx}{dy}$$

$$\therefore \frac{d\phi}{dy} = x^2 + 2xy \frac{dx}{dy}$$

$$\therefore \frac{d\phi}{dy} = 2xy \frac{dx}{dy} + x^2$$

We know,

$$d\phi = \frac{\partial \phi}{\partial x} \times dx + \frac{\partial \phi}{\partial y} \times dy \text{ -----(iii)}$$

$$\Rightarrow d\phi = 2xy \times dx + x^2 \times dy \text{ -----(iv) [From (i) & (ii)]}$$

$$\Rightarrow \frac{d\phi}{dx} = 2xy \times \frac{dx}{dx} + x^2 \frac{dy}{dx} \text{ [Dividing both sides by dx]}$$

$$\therefore \frac{d\phi}{dx} = 2xy + x^2 \frac{dy}{dx}$$

From equation (iii),

$$d\phi = \frac{\partial \phi}{\partial x} \times dx + \frac{\partial \phi}{\partial y} \times dy$$

$$\Rightarrow d\phi = 2xy \times dx + x^2 \times dy$$

$$\Rightarrow \frac{d\phi}{dy} = 2xy \frac{dx}{dy} + x^2 \frac{dy}{dy} \text{ [Dividing by dy]}$$

$$\therefore \frac{d\phi}{dy} = 2xy \frac{dx}{dy} + x^2$$

Q#29: Show that $\vec{\nabla} \phi$ is a vector perpendicular to the surface $\phi(x, y, z) = c$, where c is a constant.

Answer: Let $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$ be the position vector to any point $P(x, y, z)$ on the surface.

$\therefore d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ lies in the tangent plane to the surface at P.

Given, $\phi(x,y,z) = c$

$$\Rightarrow d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz = d(c)$$

$$\Rightarrow d\phi = \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz = 0$$

$$\Rightarrow \frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz = 0 \text{-----(i)}$$

$$\Rightarrow \left(\frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}\right) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = 0 \text{-----(ii)}$$

$$\Rightarrow \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\phi \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = 0$$

$$\Rightarrow \vec{\nabla}\phi \cdot d\vec{r} = 0 \quad \left[\because \vec{\nabla} = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} ; d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k} \right]$$

So that $\vec{\nabla}\phi$ perpendicular to $d\vec{r}$ and therefore to the surface.

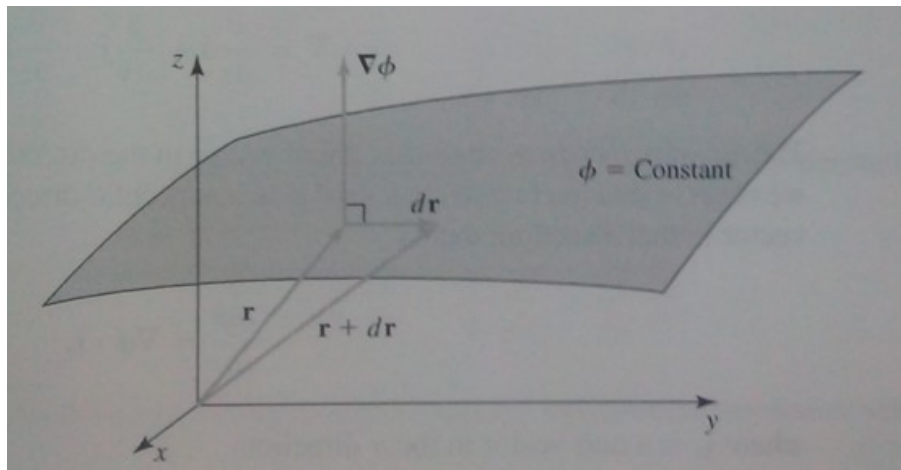
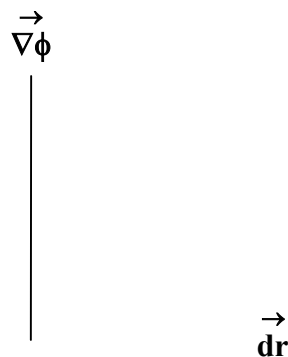


Figure # 64



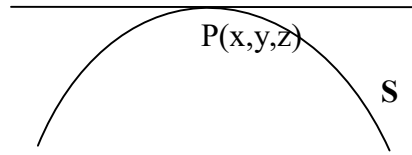


Figure # 65

It is clear that the vector $\vec{\nabla} \phi$ is perpendicular (normal) to the tangent vector $\vec{dr}$ at a point $P(x,y,z)$ that is $\vec{\nabla} \phi \cdot d\vec{r} = 0$

So we conclude that $\vec{\nabla} \phi$ is **normal (perpendicular) vector to the surface** $\phi(x,y,z) = c$ at (x,y,z) .

[N.B. We always remember that $\vec{\nabla} \phi$ is perpendicular to the tangent to the surface but not with surface directly and $\vec{\nabla} \phi$ is normal to the surface $\phi(x,y,z) = c$, Where $\vec{\nabla} \phi$ is a vector component that is $\vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$

Q# 30: Find a unit normal to the surface $x^2y + 2xz = 4$ at the point $(2,-2,3)$

Answer: Given, $\phi(x,y,z) = x^2y + 2xz = 4$

$$\begin{aligned} \therefore \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) = \hat{i} \frac{\partial}{\partial x} \phi + \hat{j} \frac{\partial}{\partial y} \phi + \hat{k} \frac{\partial}{\partial z} \phi \\ &= \hat{i} \frac{\partial}{\partial x} (x^2y + 2xz) + \hat{j} \frac{\partial}{\partial y} (x^2y + 2xz) + \hat{k} \frac{\partial}{\partial z} (x^2y + 2xz) \\ &= (2xy + 2z) \hat{i} + x^2 \hat{j} + 2x \hat{k} \\ &= (2 \times 2 \times (-2) + 2 \times 3) \hat{i} + 2^2 \hat{j} + 2 \times 2 \hat{k} \text{ at the point } (2,-2,3) \\ &= -2 \hat{i} + 4 \hat{j} + 4 \hat{k} \end{aligned}$$

$$\text{Then a unit normal to the surface} = \frac{-2 \hat{i} + 4 \hat{j} + 4 \hat{k}}{\sqrt{(-2)^2 + (4)^2 + (4)^2}} = -\frac{1}{3} \hat{i} + \frac{2}{3} \hat{j} + \frac{2}{3} \hat{k} \text{ Answer}$$

Q# 31:

Find the level surface of $F(x,y,z) = x^2 + y^2 + z^2$ passing through $(1,1,1)$. Graph the gradient at the point.

Answer: Given, $F(x,y,z) = x^2 + y^2 + z^2$

$$\therefore F(1,1,1) = 1^2 + 1^2 + 1^2 = 3$$

$$\text{Hence } F(x,y,z) = x^2 + y^2 + z^2 = 3$$

Because $F(1,1,1) = 3$, the level surface passing through $(1,1,1)$ is the sphere $x^2 + y^2 + z^2 = 3$.

The gradient of the function is

$$F(x, y, z) = x^2 + y^2 + z^2$$

$$F(x, y, z) = x^2 + y^2 + z^2$$

$$\therefore \nabla F(x, y, z) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) F$$

$$\therefore \nabla F(x, y, z) = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 + y^2 + z^2)$$

$$\therefore \nabla F(x, y, z) =$$

$$\hat{i} \frac{\partial}{\partial x} (x^2 + y^2 + z^2) + \hat{j} \frac{\partial}{\partial y} (x^2 + y^2 + z^2) + \hat{k} \frac{\partial}{\partial z} (x^2 + y^2 + z^2)$$

$$\therefore \nabla F(x, y, z) = \hat{i} \frac{\partial}{\partial x} (2x + 0 + 0) + \hat{j} \frac{\partial}{\partial y} (0 + 2y + 0) + \hat{k} \frac{\partial}{\partial z} (0 + 0 + 2z)$$

$$\therefore \nabla F(x, y, z) = \hat{i}(2x) + \hat{j}(2y) + \hat{k}(2z)$$

$$\therefore \nabla F(x, y, z) = 2x \hat{i} + 2y \hat{j} + 2z \hat{k}$$

And so, at the given point

$$\therefore \nabla F(1,1,1) = 2.1 \hat{i} + 2.1 \hat{j} + 2.1 \hat{k}$$

$$\therefore \nabla F(1,1,1) = 2 \hat{i} + 2 \hat{j} + 2 \hat{k}$$

The level surface and $\nabla F(1,1,1)$ are illustrated in figure no 63

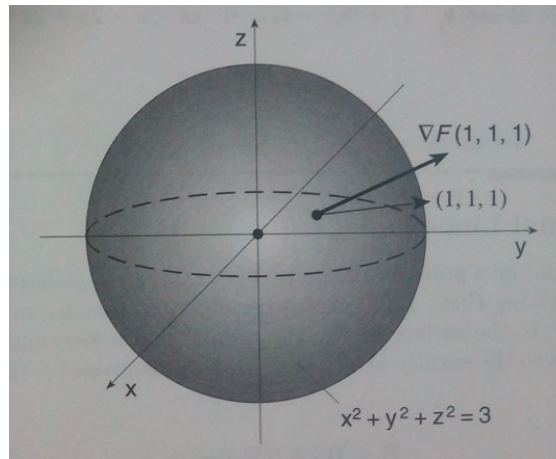


Figure # 66

Q#32: Prove that the angle between the surfaces at the point is equal to the angle between the normals to the surfaces at the point.

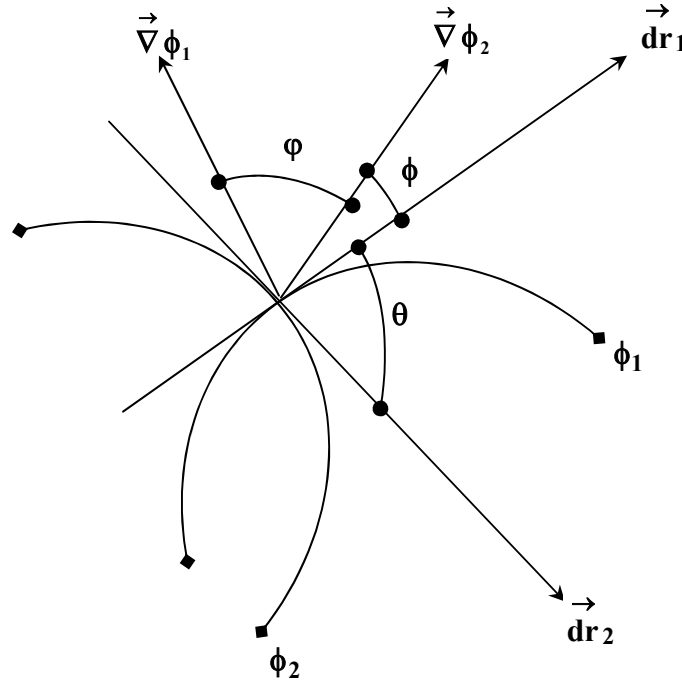


Figure # 67

Here, $\hat{\eta}_1$ is the unit vector of $\vec{\nabla} \phi_1$ and $\hat{\eta}_2$ is the unit vector of $\vec{\nabla} \phi_2$
We can write,

$$\hat{\eta}_1 = \frac{\vec{\nabla} \phi_1}{|\vec{\nabla} \phi_1|} \text{ and } \hat{\eta}_2 = \frac{\vec{\nabla} \phi_2}{|\vec{\nabla} \phi_2|}$$

We have,

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\therefore \hat{\eta}_1 \cdot \hat{\eta}_2 = |\hat{\eta}_1| |\hat{\eta}_2| \cos \phi \quad [\because \phi \text{ be the angle between the normals to the surfaces } \phi_1 \text{ and } \phi_2]$$

$$\therefore \hat{\eta}_1 \cdot \hat{\eta}_2 = 1 \cdot 1 \cos \phi \quad [\because \text{The length or magnitude of unit vector is 1}]$$

$$\Rightarrow \frac{\vec{\nabla} \phi_1}{\left| \vec{\nabla} \phi_1 \right|} \cdot \frac{\vec{\nabla} \phi_2}{\left| \vec{\nabla} \phi_2 \right|} = 1.1 \cos \varphi$$

$$\Rightarrow \vec{\nabla} \phi_1 \cdot \vec{\nabla} \phi_2 = \left| \vec{\nabla} \phi_1 \right| \left| \vec{\nabla} \phi_2 \right| \cos \varphi \text{-----(i)}$$

Again, from figure # 64

$$\varphi + \phi = 90$$

$$\pm \theta \pm \phi = \pm 90$$

$$\varphi - \theta = 0$$

$$\therefore \varphi = \theta \text{-----(ii) (Proved)}$$

Q# 33: Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$

Answer:

$$\text{Given, } z = x^2 + y^2 - 3$$

$$\Rightarrow x^2 + y^2 - z = 3$$

$$\text{Let, } \phi_1(x, y, z) = x^2 + y^2 + z^2 = 9 \text{ and } \phi_2(x, y, z) = x^2 + y^2 - z = 3$$

$$\begin{aligned} \therefore \nabla \phi_1 &= \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \phi_1 = \left(\frac{\delta \phi_1}{\delta x} \hat{i} + \frac{\delta \phi_1}{\delta y} \hat{j} + \frac{\delta \phi_1}{\delta z} \hat{k} \right) = \hat{i} \frac{\delta}{\delta x} \phi_1 + \hat{j} \frac{\delta}{\delta y} \phi_1 + \hat{k} \frac{\delta}{\delta z} \phi_1 \\ &= \hat{i} \frac{\delta}{\delta x} (x^2 + y^2 + z^2) + \hat{j} \frac{\delta}{\delta y} (x^2 + y^2 + z^2) + \hat{k} \frac{\delta}{\delta z} (x^2 + y^2 + z^2) \\ &= 2x \hat{i} + 2y \hat{j} + 2z \hat{k} \\ &= 4 \hat{i} - 2 \hat{j} + 4 \hat{k} \text{ at the point } (2, -1, 2) \end{aligned}$$

$$\text{A normal to } x^2 + y^2 + z^2 = 9 \text{ at } (2, -1, 2) \text{ is } \nabla \phi_1 = 4 \hat{i} - 2 \hat{j} + 4 \hat{k}$$

Again,

$$\phi_2(x, y, z) = x^2 + y^2 - z = 3$$

$$\begin{aligned} \therefore \nabla \phi_2 &= \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \phi_2 = \left(\frac{\delta \phi_2}{\delta x} \hat{i} + \frac{\delta \phi_2}{\delta y} \hat{j} + \frac{\delta \phi_2}{\delta z} \hat{k} \right) = \hat{i} \frac{\delta}{\delta x} \phi_2 + \hat{j} \frac{\delta}{\delta y} \phi_2 + \hat{k} \frac{\delta}{\delta z} \phi_2 \\ &= \hat{i} \frac{\delta}{\delta x} (x^2 + y^2 - z) + \hat{j} \frac{\delta}{\delta y} (x^2 + y^2 - z) + \hat{k} \frac{\delta}{\delta z} (x^2 + y^2 - z) \\ &= 2x \hat{i} + 2y \hat{j} - \hat{k} \\ &= 4 \hat{i} - 2 \hat{j} - \hat{k} \text{ at the point } (2, -1, 2) \end{aligned}$$

$$\text{A normal to } x^2 + y^2 - z = 3 \text{ at } (2, -1, 2) \text{ is } \nabla \phi_2 = 4 \hat{i} - 2 \hat{j} - \hat{k}$$

Let ϕ be the angle between the normals to the surfaces at the point $(2, -1, 2)$

Then we have,

$$\nabla\phi_1 \cdot \nabla\phi_2 = |\nabla\phi_1| |\nabla\phi_2| \cos \phi, \quad [\text{from equation no (i), page no 48}]$$

$$\Rightarrow (4\hat{i} - 2\hat{j} + 4\hat{k}) \cdot (4\hat{i} - 2\hat{j} - \hat{k}) = \left| 4\hat{i} - 2\hat{j} + 4\hat{k} \right| \left| 4\hat{i} - 2\hat{j} - \hat{k} \right| \cos \phi$$

$$\Rightarrow 16 + 4 - 4 = \sqrt{(4)^2 + (-2)^2 + (4)^2} \sqrt{(4)^2 + (-2)^2 + (-1)^2} \cos \phi$$

$$\Rightarrow 16 = 6\sqrt{21} \cos \phi$$

$$\Rightarrow \cos \phi = \frac{16}{6\sqrt{21}}$$

$$\therefore \phi = \cos^{-1}\left(\frac{16}{6\sqrt{21}}\right) \text{ Answer } [\because \phi = \theta]$$

Q# 34: Prove that $\nabla \cdot \left[\frac{\vec{f(r)}}{r} \right] = \frac{1}{r^2} \frac{d}{dr} [r^2 f(r)]$

Here $\left| \frac{\vec{r}}{r} \right| = r$

L.H.S.

$$\nabla \cdot \left[\frac{\vec{f(r)}}{r} \right]$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left[\frac{\vec{f(r)}}{r} (x\hat{i} + y\hat{j} + z\hat{k}) \right] [\because \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}]$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left(\hat{i} \frac{f(r)}{r} x + \hat{j} \frac{f(r)}{r} y + \hat{k} \frac{f(r)}{r} z \right)$$

$$= \frac{\partial}{\partial x} \frac{f(r)}{r} x + \frac{\partial}{\partial y} \frac{f(r)}{r} y + \frac{\partial}{\partial z} \frac{f(r)}{r} z \text{-----(i)}$$

$$[\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1]$$

$$\text{Now, } \frac{\partial}{\partial x} \frac{f(r)}{r} x$$

$$= \frac{\partial}{\partial x} \left\{ \frac{f(r)}{r} x \right\}$$

$$= \frac{f(r)}{r} \frac{\partial}{\partial x} (x) + x \frac{\partial}{\partial x} \frac{f(r)}{r}$$

$$= \frac{f(r)}{r} \cdot 1 + x \frac{\partial}{\partial x} \frac{f(r)}{r}$$

$$= \frac{f(r)}{r} + x \frac{\partial}{\partial x} \frac{f(r)}{r}$$

$$= \frac{f(r)}{r} + x \frac{\partial}{\partial x} \{f(r)r^{-1}\}$$

$$\begin{aligned}
&= \frac{f(r)}{r} + x[f(r)\frac{\delta}{\delta x}(r^{-1}) + r^{-1}\frac{\delta}{\delta x}\{f(r)\}] \\
&= \frac{f(r)}{r} + x[f(r)(-1)(r^{-2})\frac{\delta r}{\delta x} + r^{-1}\frac{\delta}{\delta x}\{f(r)\}] \\
&= \frac{f(r)}{r} + x[f(r)(-1)(r^{-2})\frac{\delta r}{\delta x} + r^{-1}\{f'(r)\}\frac{\delta r}{\delta x}] \\
&= \frac{f(r)}{r} + x[-f(r)(r^{-2})\frac{\delta r}{\delta x} + r^{-1}\{f'(r)\}\frac{\delta r}{\delta x}] \\
&= \frac{f(r)}{r} + x[r^{-1}\{f'(r)\}\frac{\delta r}{\delta x} - f(r)(r^{-2})\frac{\delta r}{\delta x}] \\
&= \frac{f(r)}{r} + x[\frac{1}{r}\{f'(r)\}\frac{\delta r}{\delta x} - f(r)\frac{1}{r^2}\frac{\delta r}{\delta x}] \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{\delta r}{\delta x} \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{\delta}{\delta x}(x^2 + y^2 + z^2)^{1/2} \\
&[\because \left| \frac{\vec{r}}{r} \right| = r = \sqrt{x^2 + y^2 + z^2} = (x^2 + y^2 + z^2)^{1/2}] \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{1}{2}(x^2 + y^2 + z^2)^{-1/2} \cdot \frac{\delta}{\delta x}(x^2 + y^2 + z^2) \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{1}{2}(x^2 + y^2 + z^2)^{-1/2} \cdot (2x) \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}](x^2 + y^2 + z^2)^{-1/2} \cdot (x) \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{x}{(x^2 + y^2 + z^2)^{1/2}} \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{x}{\sqrt{x^2 + y^2 + z^2}} \\
&= \frac{f(r)}{r} + x[\frac{f'(r)}{r} - \frac{f(r)}{r^2}]\frac{x}{r} \\
&= \frac{f(r)}{r} + \frac{x^2 f'(r)}{r^2} - \frac{x^2 f(r)}{r^3} \text{-----(ii)}
\end{aligned}$$

Similarly,

$$\frac{\delta}{\delta y} \frac{f(r)}{r} y = \frac{f(r)}{r} + \frac{y^2 f'(r)}{r^2} - \frac{y^2 f(r)}{r^3} \text{-----(iii)}$$

and

$$\frac{\delta}{\delta z} \frac{f(r)}{r} z = \frac{f(r)}{r} + \frac{z^2 f'(r)}{r^2} - \frac{z^2 f(r)}{r^3} \text{-----(iv)}$$

Putting the value of (ii), (iii) and (iv) in (i)

$$\begin{aligned} \nabla \cdot \left[\frac{f(r)}{r} \vec{r} \right] &= \frac{\delta}{\delta x} \frac{f(r)}{r} x + \frac{\delta}{\delta y} \frac{f(r)}{r} y + \frac{\delta}{\delta z} \frac{f(r)}{r} z \\ &= \frac{f(r)}{r} + \frac{x^2 f'(r)}{r^2} - \frac{x^2 f(r)}{r^3} + \frac{f(r)}{r} + \frac{y^2 f'(r)}{r^2} - \frac{y^2 f(r)}{r^3} + \frac{f(r)}{r} + \frac{z^2 f'(r)}{r^2} - \frac{z^2 f(r)}{r^3} \\ &= 3 \frac{f(r)}{r} + \frac{x^2 f'(r)}{r^2} - \frac{x^2 f(r)}{r^3} + \frac{y^2 f'(r)}{r^2} - \frac{y^2 f(r)}{r^3} + \frac{z^2 f'(r)}{r^2} - \frac{z^2 f(r)}{r^3} \\ &= 3 \frac{f(r)}{r} + \frac{f'(r)}{r^2} (x^2 + y^2 + z^2) - \frac{f(r)}{r^3} (x^2 + y^2 + z^2) \\ &= 3 \frac{f(r)}{r} + \frac{f'(r)}{r^2} r^2 - \frac{f(r)}{r^3} r^2 \\ &= 3 \frac{f(r)}{r} + f'(r) - \frac{f(r)}{r} \\ &= 2 \frac{f(r)}{r} + f'(r) \\ &= \frac{1}{r^2} \cdot r^2 \left[2 \frac{f(r)}{r} + f'(r) \right] \\ &= \frac{1}{r^2} [2rf(r) + r^2 f'(r)] \\ &= \frac{1}{r^2} \frac{d}{dr} [r^2 f(r)] \quad [\because \frac{d}{dr} [r^2 f(r)] = 2rf(r) + r^2 f'(r)] \\ &\text{(Proved)} \end{aligned}$$

Directional derivatives

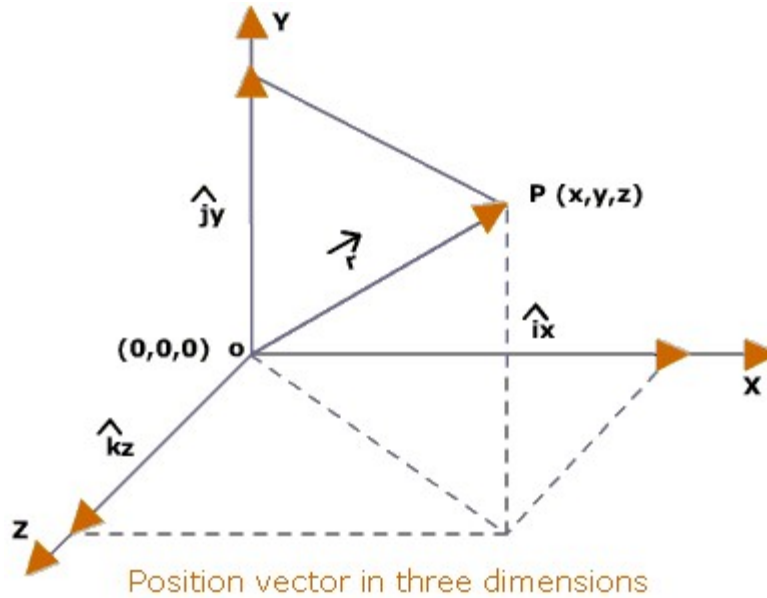


Figure # 68

Let $\vec{OP}$ is a position vector $\vec{r}$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $d\vec{r}$ is a small displacement corresponding to changes dx, dy, dz in x, y, z respectively, then

$$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k} \dots\dots\dots(i)$$

If $\phi(x, y, z)$ is a scalar function at P, then the *gradient* of ϕ

$$\text{grad } \phi = \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \dots\dots\dots(ii)$$

$$\begin{aligned} \text{Then } \text{grad } \phi \cdot d\vec{r} &= \vec{\nabla} \phi \cdot d\vec{r} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) \\ &= \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz \\ &= d\phi \\ &= \text{The total differential } d\phi \text{ of } \phi \end{aligned}$$

$$\text{grad } \phi \cdot d\vec{r} = d\phi$$

$$d\phi = \text{grad } \phi \cdot d\vec{r}$$

$$d\phi = \vec{\nabla} \phi \cdot d\vec{r} \dots\dots\dots(iii)$$

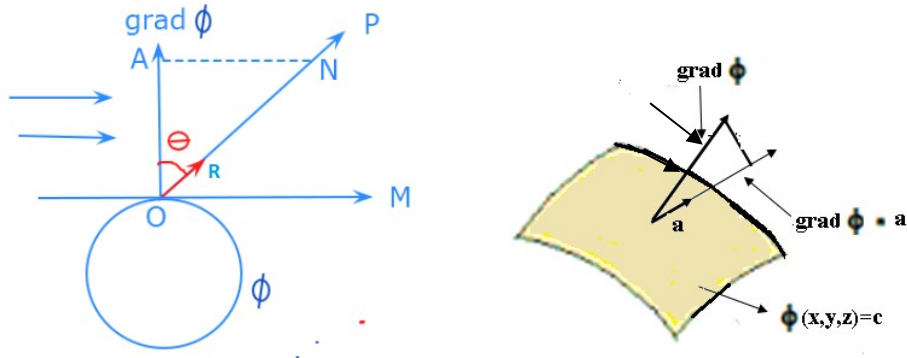


Figure # 69

We have just established that

$$d\phi = dr \cdot \text{grad } \phi$$

A

If ds is the small element of arc between $P(r)$ and $Q(r+dr)$ then $ds = \left| \frac{d\vec{r}}{ds} \right|$

$$\frac{d\vec{r}}{ds} = \frac{d\vec{r}}{\left| d\vec{r} \right|}$$

and $\frac{d\vec{r}}{ds}$ is thus a unit vector in the direction of $d\vec{r}$.

$$\therefore \frac{d\phi}{ds} = \frac{d\vec{r}}{ds} \cdot \text{Grad } \phi$$

If we denote this unit vector by $\hat{a}$, i.e. $\frac{d\vec{r}}{ds} = \hat{a}$, the result becomes

$$\frac{d\phi}{ds} = \hat{a} \cdot \text{Grad } \phi$$

$\frac{d\phi}{ds}$ is the projection of $\text{grad } \phi$ on the unit vector $\hat{a}$ is called the directional derivative of ϕ in the direction of $\hat{a}$. It gives the rate of change of ϕ with distance measured in the direction of $\hat{a}$ and $\frac{d\phi}{ds} = \hat{a} \cdot \text{Grad } \phi$ will be a maximum when $\hat{a}$ and $\text{grad } \phi$ have the

same direction, since then, $\hat{a} \cdot \text{grad } \phi = \left| \hat{a} \right| \left| \text{grad } \phi \right| \cos \theta$ and θ will be zero

Thus direction of $\text{grad } \phi$ gives the direction in which the maximum rate of change of ϕ occurs.

Q# 35: Find the directional derivative of the function $\phi = x^2z + 2xy^2 + yz^2$ at the point of (1, 2, -1) in the direction of the vector $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$.

We start off with $\phi = x^2z + 2xy^2 + yz^2$

$$\therefore \nabla\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\phi$$

$$\therefore \nabla\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)(x^2z + 2xy^2 + yz^2)$$

$$\therefore \nabla\phi =$$

$$\hat{i}\frac{\partial}{\partial x}(x^2z + 2xy^2 + yz^2) + \hat{j}\frac{\partial}{\partial y}(x^2z + 2xy^2 + yz^2) + \hat{k}\frac{\partial}{\partial z}(x^2z + 2xy^2 + yz^2)$$

$$\therefore \nabla\phi = \hat{i}(2xz + 2.1.y^2 + 0) + \hat{j}(0 + 2x.2y + 1.z^2) + \hat{k}(x^2.1 + 0 + y.2z)$$

$$\therefore \nabla\phi = \hat{i}(2xz + 2y^2) + \hat{j}(4xy + z^2) + \hat{k}(x^2 + 2yz)$$

At the point (1, 2, -1)

$$\nabla\phi = \hat{i}(2xz + 2y^2) + \hat{j}(4xy + z^2) + \hat{k}(x^2 + 2yz)$$

$$\therefore \nabla\phi = \hat{i}[2.1.(-1) + 2(2^2)] + \hat{j}[4.1.2 + (-1)^2] + \hat{k}[(1^2 + 2.2.(-1))]$$

$$\therefore \nabla\phi = \hat{i}[-2 + 8] + \hat{j}[8 + 1] + \hat{k}[1 - 4]$$

$$\therefore \nabla\phi = \hat{i}[6] + \hat{j}[9] + \hat{k}(-3)$$

$$\therefore \nabla\phi = 6\hat{i} + 9\hat{j} - 3\hat{k}$$

Next we have to find out the unit vector $\hat{a}$ where $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$

$$\therefore \hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|2\hat{i} + 3\hat{j} - 4\hat{k}|}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{\sqrt{4 + 9 + 16}}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{\sqrt{29}}$$

$$\text{Hence } \frac{d\phi}{ds} = \hat{a} \cdot \nabla\phi$$

$$\frac{d\phi}{ds} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|29|} \cdot (6\hat{i} + 9\hat{j} - 3\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|29|} (2\hat{i} + 3\hat{j} - 4\hat{k}) \cdot (6\hat{i} + 9\hat{j} - 3\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|29|} (12 + 27 + 12)$$

$$\frac{d\phi}{ds} = \frac{51}{|29|} \text{ Answer}$$

Q#36: Find the directional derivative of the function $\phi = x^2y + y^2z + z^2x$ at the point of $(1, -1, 2)$ in the direction of the vector $\vec{A} = 4\hat{i} + 2\hat{j} - 5\hat{k}$.

We have $\phi = x^2y + y^2z + z^2x$

$$\therefore \nabla\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\phi$$

$$\therefore \nabla\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)(x^2y + y^2z + z^2x)$$

$$\therefore \nabla\phi = \hat{i}\frac{\partial}{\partial x}(x^2y + y^2z + z^2x) + \hat{j}\frac{\partial}{\partial y}(x^2y + y^2z + z^2x) + \hat{k}\frac{\partial}{\partial z}(x^2y + y^2z + z^2x)$$

$$\therefore \nabla\phi = \hat{i}(2xy + 0 + z^2 \cdot 1) + \hat{j}(x^2 \cdot 1 + 2yz + 0) + \hat{k}(0 + y^2 \cdot 1 + 2zx)$$

$$\therefore \nabla\phi = \hat{i}(2xy + z^2) + \hat{j}(x^2 + 2yz) + \hat{k}(y^2 + 2zx)$$

At the point $(1, -1, 2)$

$$\therefore \nabla\phi = \hat{i}(2xy + z^2) + \hat{j}(x^2 + 2yz) + \hat{k}(y^2 + 2zx)$$

$$\therefore \nabla\phi = \hat{i}[2 \cdot 1 \cdot (-1) + 2^2] + \hat{j}[1^2 + 2(-1) \cdot 2] + \hat{k}[(-1)^2 + 2 \cdot 2 \cdot 1]$$

$$\therefore \nabla\phi = \hat{i}[-2 + 4] + \hat{j}[1 - 4] + \hat{k}[1 + 4]$$

$$\therefore \nabla\phi = 2\hat{i} - 3\hat{j} + 5\hat{k}$$

Next we have to find out the unit vector $\hat{a}$ where $\vec{A} = 4\hat{i} + 2\hat{j} - 5\hat{k}$

$$\therefore \hat{a} = \frac{\vec{A}}{|\vec{A}|}$$

$$\therefore \hat{a} = \frac{4\hat{i} + 2\hat{j} - 5\hat{k}}{|4^2 + 2^2 + (-5)^2|}$$

$$\therefore \hat{a} = \frac{4\hat{i} + 2\hat{j} - 5\hat{k}}{|16 + 4 + 25|}$$

$$\therefore \hat{a} = \frac{4\hat{i} + 2\hat{j} - 5\hat{k}}{|45|}$$

$$\text{Hence } \frac{d\phi}{ds} = \hat{a} \cdot \vec{\nabla} \phi$$

$$\frac{d\phi}{ds} = \frac{(4\hat{i} + 2\hat{j} - 5\hat{k})}{|45|} \cdot (2\hat{i} - 3\hat{j} + 5\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|45|} (4\hat{i} + 2\hat{j} - 5\hat{k}) \cdot (2\hat{i} - 3\hat{j} + 5\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|45|} (8 - 6 - 25)$$

$$\frac{d\phi}{ds} = \frac{1}{|45|} (-23) \text{ Answer}$$

Q#37: Find the directional derivative of the function $\phi = (x, y, z) = x^2 - y^2 + 2z^2$ at the point of (1,2,3) in the direction of the vector $\vec{A} = 4\hat{i} - 2\hat{j} + \hat{k}$.

Answer: Let, $\phi(x, y, z) = x^2 - y^2 + 2z^2$

$$\frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x} (x^2 - y^2 + 2z^2)$$

$$\frac{\partial \phi}{\partial x} = (2x - 0 + 0)$$

$$\frac{\partial \phi}{\partial x} = 2x$$

$$\phi = (x, y, z) = x^2 - y^2 + 2z^2$$

$$\frac{\partial \phi}{\partial y} = \frac{\partial}{\partial y} (x^2 - y^2 + 2z^2)$$

$$\frac{\partial \phi}{\partial y} = (0 - 2y + 0)$$

$$\frac{\partial \phi}{\partial y} = -2y$$

$$\phi(x, y, z) = x^2 - y^2 + 2z^2$$

$$\frac{\partial \phi}{\partial z} = \frac{\partial}{\partial z} (x^2 - y^2 + 2z^2)$$

$$\frac{\partial \phi}{\partial z} = (0 - 0 + 4z)$$

$$\frac{\partial \phi}{\partial z} = 4z$$

$$\phi(x, y, z) = x^2 - y^2 + 2z^2$$

$$\text{grad } \phi = \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi$$

$$\text{grad } \phi = \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 - y^2 + 2z^2)$$

$$\text{grad } \phi = \vec{\nabla} \phi = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (x^2 - y^2 + 2z^2)$$

$$\text{grad } \phi = \vec{\nabla} \phi = \hat{i} \frac{\partial}{\partial x} (x^2 - y^2 + 2z^2) + \hat{j} \frac{\partial}{\partial y} (x^2 - y^2 + 2z^2) + \hat{k} \frac{\partial}{\partial z} (x^2 - y^2 + 2z^2)$$

$$\text{grad } \phi = \vec{\nabla} \phi = \hat{i} 2x + \hat{j} (-2y) + \hat{k} 4z$$

$$\text{grad } \phi = \vec{\nabla} \phi(1,2,3) = \hat{i} (2 \times 1) + \hat{j} (-2 \times 2) + \hat{k} (4 \times 3)$$

$$\text{grad } \phi = \vec{\nabla} \phi(1,2,3) = 2\hat{i} - 4\hat{j} + 12\hat{k}$$

Given ,

$$\vec{A} = 4\hat{i} - 2\hat{j} + \hat{k}$$

$$\hat{A} = \frac{\left| \vec{A} \right|}{\left| \vec{A} \right|} = \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{(4)^2 + (-2)^2 + (1)^2}} = \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{16 + 4 + 1}} = \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{21}}$$

∴ The directional derivative of the function $\phi = (x, y, z) = x^2 - y^2 + 2z^2$ at the point of

$$(1,2,3) \text{ in the direction of the vector } \vec{A} = \vec{\nabla} \phi. \hat{A} = (2\hat{i} - 4\hat{j} + 12\hat{k}). \frac{4\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{21}}$$

$$= \frac{8 + 8 + 12}{\sqrt{21}} = \frac{28}{\sqrt{21}}$$

Q# 38 Suppose that over a certain region of space the electrical potential V is given by

$$\phi(x, y, z) = 5x^2 - 3xy + xyz$$

- (i) Find the rate of change (derivative) of the potential at P(3,4,3) in the direction of the

$$\text{vector } \vec{v} = \hat{i} + \hat{j} - \hat{k}$$

- (ii) In which direction does ϕ changes most rapidly at P?

- (iii) What is the maximum rate of change at P?

$$\begin{aligned} \text{Answer: grad } \phi &= \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi \\ &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (5x^2 - 3xy + xyz) \end{aligned}$$

$$\begin{aligned}
&= (\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z})(5x^2 - 3xy + xyz) \\
&= \hat{i} \frac{\partial}{\partial x}(5x^2 - 3xy + xyz) + \hat{j} \frac{\partial}{\partial y}(5x^2 - 3xy + xyz) + \hat{k} \frac{\partial}{\partial z}(5x^2 - 3xy + xyz) \\
&= \hat{i}(10x - 3y + yz) + \hat{j}(-3x + xz) + \hat{k}(xy)
\end{aligned}$$

At P (3, 4, 3),

$$\begin{aligned}
\vec{\nabla} \phi &= \hat{i}(10x - 3y + yz) + \hat{j}(-3x + xz) + \hat{k}(xy) \\
\vec{\nabla} \phi &= \hat{i}(10 \times 3 - 3 \times 4 + 4 \times 3) + \hat{j}(-3 \times 3 + 3 \times 3) + \hat{k}(3 \times 4) \\
\vec{\nabla} \phi &= \hat{i}(30 - 12 + 12) + \hat{j}(-9 + 9) + \hat{k}(12) \\
\vec{\nabla} \phi &= \hat{i}(30) + 12\hat{k}
\end{aligned}$$

Given, $\vec{v} = \hat{i} + \hat{j} - \hat{k}$; the unit vector of $\vec{v} = \hat{a} = \frac{\vec{v}}{|\vec{v}|} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{1^2 + 1^2 + (-1)^2}} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$

i)

$$\begin{aligned}
\vec{\nabla} \phi \cdot \hat{a} &= [\hat{i}(30) + 12\hat{k}] \cdot \left[\frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}} \right] \\
\vec{\nabla} \phi \cdot \hat{a} &= [\hat{i}(30) + 12\hat{k}] \cdot \left[\frac{\hat{i}}{\sqrt{3}} + \frac{\hat{j}}{\sqrt{3}} - \frac{\hat{k}}{\sqrt{3}} \right] \\
\vec{\nabla} \phi \cdot \hat{a} &= \left[\frac{30}{\sqrt{3}} - \frac{12}{\sqrt{3}} \right] \\
\vec{\nabla} \phi \cdot \hat{a} &= \frac{30 - 12}{\sqrt{3}} = \frac{18}{\sqrt{3}} = \frac{18\sqrt{3}}{3} = 6\sqrt{3}
\end{aligned}$$

ii)

$$\vec{\nabla} \phi = \hat{i}(30) + 12\hat{k}$$

iii)

$$|\vec{\nabla} \phi| = \sqrt{(30)^2 + (12)^2} = \sqrt{900 + 144} = \sqrt{1044}$$

Q# 39: If $\vec{V}(x, y, z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ Find divergence of $\vec{V}$ that is $\text{div } \vec{V}$

Answer: $\text{div } \vec{V} = \nabla \cdot \vec{V} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (xz\hat{i} + xyz\hat{j} - y^2\hat{k})$

$$\begin{aligned}
&= \frac{\partial}{\partial x}(xz) + \frac{\partial}{\partial y}(xyz) - \frac{\partial}{\partial z}(y^2) \quad [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{k} \cdot \hat{i} = 0 \text{ etc.}] \\
&= z + xz \quad \text{Answer}
\end{aligned}$$

Q# 40: Let $\vec{V}$ be a constant vector field. Show that $\text{div } \vec{V} = 0$

Answer: Let, $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$, where a, b, c are constants, Then

$$\begin{aligned}\text{div } \vec{V} &= \nabla \cdot \vec{V} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (a\hat{i} + b\hat{j} + c\hat{k}) \\ &= \frac{\partial}{\partial x}(a) + \frac{\partial}{\partial y}(b) + \frac{\partial}{\partial z}(c) \quad [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{k} \cdot \hat{i} = 0 \text{ etc.}] \\ &= 0\end{aligned}$$

Q# 41: what is solenoidal?

Answer: If $\vec{A}$ is solenoidal then $\nabla \cdot \vec{A} = 0$

Q# 42: Show that the vector field $\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{x^2 + y^2}}$ is a “sink field”. Plot and give a physical interpretation. [Here $\hat{V}$ is a unit vector; Since it is divided by its length]

Answer: given,

$$\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{x^2 + y^2}}$$

$$\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{(-x)^2 + (-y)^2}}$$

$$\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{x^2 + y^2}}$$

$$\vec{v} = \frac{-x}{\sqrt{x^2 + y^2}} \hat{i} + \frac{-y}{\sqrt{x^2 + y^2}} \hat{j}$$

$$\therefore \text{div } \vec{v} \equiv \nabla \cdot \vec{v} \equiv \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \left(\frac{-x}{\sqrt{x^2 + y^2}} \hat{i} + \frac{-y}{\sqrt{x^2 + y^2}} \hat{j} + 0 \cdot \hat{k} \right)$$

$$\text{div } \vec{v} \equiv \nabla \cdot \vec{v} \equiv \frac{\partial}{\partial x} \left(\frac{-x}{\sqrt{x^2 + y^2}} \right) + \frac{\partial}{\partial y} \left(\frac{-y}{\sqrt{x^2 + y^2}} \right)$$

$$\text{div } \vec{v} \equiv \nabla \cdot \vec{v} \equiv \frac{\partial}{\partial x} \left(\frac{-x}{\sqrt{x^2 + y^2}} \right) + \frac{\partial}{\partial y} \left(\frac{-y}{\sqrt{x^2 + y^2}} \right)$$

$$\text{div } \vec{v} \equiv \nabla \cdot \vec{v} \equiv \frac{(\sqrt{x^2 + y^2}) \frac{\partial}{\partial x}(-x) - (-x) \frac{\partial}{\partial x}(\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} + \frac{(\sqrt{x^2 + y^2}) \frac{\partial}{\partial y}(-y) - (-y) \frac{\partial}{\partial y}(\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2}$$

$$\begin{aligned}
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{(\sqrt{x^2 + y^2})(-1) + x \frac{\delta}{\delta x}(\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} + \frac{(\sqrt{x^2 + y^2})(-1) + y \frac{\delta}{\delta y}(\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + x \frac{\delta}{\delta x}(\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} + \frac{-(\sqrt{x^2 + y^2}) + y \frac{\delta}{\delta y}(\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + x \frac{\delta}{\delta x}(x^2 + y^2)^{1/2}}{(\sqrt{x^2 + y^2})^2} + \frac{-(\sqrt{x^2 + y^2}) + y \frac{\delta}{\delta y}(x^2 + y^2)^{1/2}}{(\sqrt{x^2 + y^2})^2} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + x \frac{1}{2}(x^2 + y^2)^{\frac{1}{2}-1} \cdot \frac{\delta}{\delta x}(x^2 + y^2)}{(\sqrt{x^2 + y^2})^2} + \frac{-(\sqrt{x^2 + y^2}) + y \frac{1}{2}(x^2 + y^2)^{\frac{1}{2}-1} \cdot \frac{\delta}{\delta y}(x^2 + y^2)}{(\sqrt{x^2 + y^2})^2} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + x \frac{1}{2}(x^2 + y^2)^{-\frac{1}{2}} \cdot (2x)}{(\sqrt{x^2 + y^2})^2} + \frac{-(\sqrt{x^2 + y^2}) + y \frac{1}{2}(x^2 + y^2)^{-\frac{1}{2}} \cdot (2y)}{(\sqrt{x^2 + y^2})^2} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + x^2(x^2 + y^2)^{-\frac{1}{2}}}{(\sqrt{x^2 + y^2})^2} + \frac{-(\sqrt{x^2 + y^2}) + y^2(x^2 + y^2)^{-\frac{1}{2}}}{(\sqrt{x^2 + y^2})^2} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + \frac{x^2}{(x^2 + y^2)^{\frac{1}{2}}}}{x^2 + y^2} + \frac{-(\sqrt{x^2 + y^2}) + \frac{y^2}{(x^2 + y^2)^{\frac{1}{2}}}}{x^2 + y^2} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(\sqrt{x^2 + y^2}) + \frac{x^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2} + \frac{-(\sqrt{x^2 + y^2}) + \frac{y^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{\frac{-x^2 - y^2 + x^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2} + \frac{\frac{-x^2 - y^2 + y^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{\frac{-y^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2} + \frac{\frac{-x^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-y^2}{(x^2 + y^2)(\sqrt{x^2 + y^2})} + \frac{-x^2}{(x^2 + y^2)(\sqrt{x^2 + y^2})} \\
\text{div } \vec{v} &\equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-y^2 - x^2}{(x^2 + y^2)(\sqrt{x^2 + y^2})}
\end{aligned}$$

$$\text{div } \vec{v} \equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-(x^2 + y^2)}{(x^2 + y^2)(\sqrt{x^2 + y^2})}$$

$$\text{div } \vec{v} \equiv \vec{\nabla} \cdot \vec{v} \equiv \frac{-1}{\sqrt{x^2 + y^2}} < 0$$

So, $\vec{v}$ a “sink field”

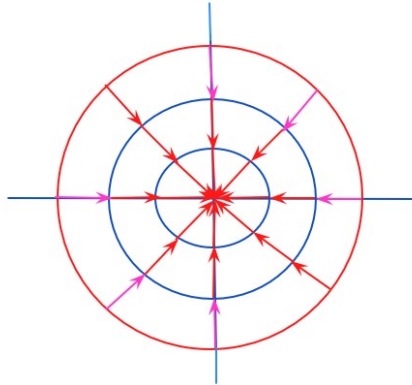


Figure # 70

Q# 43: Let $\vec{V}$ be a constant vector field. Show that $\text{curl } \vec{V} = 0$

Answer Let, $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$, where a, b, c are constants, Then

$$\text{curl } \vec{V} = \nabla \times \vec{V} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \times (a\hat{i} + b\hat{j} + c\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a & b & c \end{vmatrix}$$

$$= \hat{i} \left(\frac{\partial c}{\partial y} - \frac{\partial b}{\partial z} \right) - \hat{j} \left(\frac{\partial c}{\partial x} - \frac{\partial a}{\partial z} \right) + \hat{k} \left(\frac{\partial b}{\partial x} - \frac{\partial a}{\partial y} \right)$$

$$= \hat{i}(0 - 0) - \hat{j}(0 - 0) + \hat{k}(0 - 0)$$

$$= 0$$

Q# 44: If $\vec{v}(x, y, z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ Find $\text{curl } \vec{V}$

Answer: The curl of a vector field $\vec{v}(x, y, z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ is defined by

$$\begin{aligned}
\nabla \times \vec{v} &= \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \times (xz \hat{i} + xyz \hat{j} - y^2 \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ xz & xyz & -y^2 \end{vmatrix} \\
&= \hat{i} \left[\frac{\delta}{\delta y} (-y^2) - \frac{\delta}{\delta z} (xyz) \right] - \hat{j} \left[\frac{\delta}{\delta x} (-y^2) - \frac{\delta}{\delta z} (xz) \right] + \hat{k} \left[\frac{\delta}{\delta x} (xyz) - \frac{\delta}{\delta y} (xz) \right] \\
&= \hat{i} [-2y - xy] - \hat{j} [0 - x] + \hat{k} [yz - 0] \\
&= -[2y + xy] \hat{i} + \hat{j} x + \hat{k} yz \\
&= -[2y + xy] \hat{i} + x \hat{j} + yz \hat{k}
\end{aligned}$$

Q# 45: What is irrotational Field or conservative vector field?

A vector field $\vec{V}$ for which the curl vanishes, that is: $\vec{V} \times \vec{V} = 0$

Q# 46: Determine $\vec{F}$ is a conservative vector field or not where

$$\vec{F} = x^2y \hat{i} + xyz \hat{j} - x^2y^2 \hat{k}$$

Answer:

So all that we need to do is compute the curl and see if we get the zero vector or not.

The curl of a vector field $\vec{F} = x^2y \hat{i} + xyz \hat{j} - x^2y^2 \hat{k}$ is defined by

$$\begin{aligned}
\nabla \times \vec{F} &= \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \times (x^2y \hat{i} + xyz \hat{j} - x^2y^2 \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ x^2y & xyz & -x^2y^2 \end{vmatrix} \\
&= \hat{i} \left[\frac{\delta}{\delta y} (-x^2y^2) - \frac{\delta}{\delta z} (xyz) \right] - \hat{j} \left[\frac{\delta}{\delta x} (-x^2y^2) - \frac{\delta}{\delta z} (x^2y) \right] \\
&\quad + \hat{k} \left[\frac{\delta}{\delta x} (xyz) - \frac{\delta}{\delta y} (x^2y) \right] \\
&= \hat{i} [-2x^2y - xy] - \hat{j} [-2xy^2 - 0] + \hat{k} [yz - x^2] \\
&\neq 0
\end{aligned}$$

So, the curl isn't the zero vectors and so this vector field is not conservative.

Q# 47: If $\vec{A} = (xz^3 \hat{i} - 2x^2yz \hat{j} + 2yz^4 \hat{k})$, find $\nabla \times \vec{A}$ (or curl A) at the point (1, -1, 1).

Answer:

$$\nabla \times \vec{A} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \times (xz^3 \hat{i} - 2x^2yz \hat{j} + 2yz^4 \hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (2yz^4) - \frac{\partial}{\partial z} (-2x^2yz) \right] - \hat{j} \left[\frac{\partial}{\partial x} (2yz^4) - \frac{\partial}{\partial z} (xz^3) \right] + \hat{k} \left[\frac{\partial}{\partial x} (-2x^2yz) - \frac{\partial}{\partial y} (xz^3) \right]$$

$$= \hat{i} [2z^4 + 2x^2y] - \hat{j} [0 - 3xz^2] + \hat{k} [-4xyz - 0]$$

$$= \hat{i} [2.1^4 + 2.1^2 \cdot (-1)] - \hat{j} [0 - 3.1.1^2] + \hat{k} [-4.1.(-1).1 - 0]$$

$$= \hat{i} [2 - 2] - \hat{j} [0 - 3] + \hat{k} [4]$$

$$= 3\hat{j} + 4\hat{k}$$

Q# 48: Prove that; $\text{curl}(\phi \vec{F}) = \text{grad} \phi \times \vec{F}$; if $\vec{F}$ is irrotational and $\phi(x, y, z)$ is a Scalar function.

Answer: Let, $\vec{F} = F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}$

$$\therefore \text{curl}(\phi \vec{F}) = \nabla \times (\phi \vec{F})$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times [\phi (F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k})]$$

$$= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times (\phi F_1 \hat{i} + \phi F_2 \hat{j} + \phi F_3 \hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \phi F_1 & \phi F_2 & \phi F_3 \end{vmatrix}$$

$$= \hat{i} \left[\frac{\partial}{\partial y} (\phi F_3) - \frac{\partial}{\partial z} (\phi F_2) \right] - \hat{j} \left[\frac{\partial}{\partial x} (\phi F_3) - \frac{\partial}{\partial z} (\phi F_1) \right] + \hat{k} \left[\frac{\partial}{\partial x} (\phi F_2) - \frac{\partial}{\partial y} (\phi F_1) \right]$$

$$= \hat{i} \left[\phi \frac{\partial}{\partial y} (F_3) + F_3 \frac{\partial \phi}{\partial y} - \phi \frac{\partial}{\partial z} (F_2) - F_2 \frac{\partial \phi}{\partial z} \right] - \hat{j} \left[\phi \frac{\partial}{\partial x} (F_3) + F_3 \frac{\partial \phi}{\partial x} - \phi \frac{\partial}{\partial z} (F_1) - F_1 \frac{\partial \phi}{\partial z} \right]$$

$$+ \hat{k} \left[\phi \frac{\partial}{\partial x} (F_2) + F_2 \frac{\partial \phi}{\partial x} - \phi \frac{\partial}{\partial y} (F_1) - F_1 \frac{\partial \phi}{\partial y} \right]$$

$$\begin{aligned}
&= \phi \left[\hat{i} \left\{ \frac{\delta}{\delta y} (F_3) - \frac{\delta}{\delta z} (F_2) \right\} - \hat{j} \left\{ \frac{\delta}{\delta x} (F_3) - \frac{\delta}{\delta z} (F_1) \right\} + \hat{k} \left\{ \frac{\delta}{\delta x} (F_2) - \frac{\delta}{\delta y} (F_1) \right\} \right] \\
&+ \hat{i} \left[F_3 \frac{\delta \phi}{\delta y} - F_2 \frac{\delta \phi}{\delta z} \right] - \hat{j} \left[F_3 \frac{\delta \phi}{\delta x} - F_1 \frac{\delta \phi}{\delta z} \right] + \hat{k} \left[F_2 \frac{\delta \phi}{\delta x} - F_1 \frac{\delta \phi}{\delta y} \right] \\
&= \phi \left| \begin{array}{ccc} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ F_1 & F_2 & F_3 \end{array} \right| + \left| \begin{array}{ccc} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta \phi}{\delta x} & \frac{\delta \phi}{\delta y} & \frac{\delta \phi}{\delta z} \\ F_1 & F_2 & F_3 \end{array} \right| \\
&= \phi (\nabla \times \mathbf{F}) + \nabla \phi \times \mathbf{F} \\
&= \phi \cdot \mathbf{0} + \nabla \phi \times \mathbf{F} \quad [\because (\nabla \times \mathbf{F}) = \mathbf{0} \text{ for irrotational}] \\
&= \nabla \phi \times \mathbf{F} \\
&= \text{grad} \phi \times \mathbf{F} \text{ (Proved)}
\end{aligned}$$

Q# 49: Prove that $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$

Answer: Let $\vec{a}, \vec{b}, \vec{c}$ are three vectors

Then $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$

$\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$

$\vec{c} = c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}$

$$\begin{aligned}
(\vec{b} \times \vec{c}) &= (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \times (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}) \\
&= \mathbf{0} + b_1 c_2 \hat{k} + b_1 c_3 (-\hat{j}) + b_2 c_1 (-\hat{k}) + \mathbf{0} + b_2 c_3 \hat{i} + b_3 c_1 \hat{j} + b_3 c_2 (-\hat{i}) + \mathbf{0} \\
&= b_1 c_2 \hat{k} - b_1 c_3 \hat{j} - b_2 c_1 \hat{k} + b_2 c_3 \hat{i} + b_3 c_1 \hat{j} - b_3 c_2 \hat{i} \\
&= \hat{i} (b_2 c_3 - b_3 c_2) + \hat{j} (b_3 c_1 - b_1 c_3) + \hat{k} (b_1 c_2 - b_2 c_1)
\end{aligned}$$

Now,

$\vec{a} \times (\vec{b} \times \vec{c})$

$$\begin{aligned}
&= (a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \times \{ \hat{i} (b_2 c_3 - b_3 c_2) + \hat{j} (b_3 c_1 - b_1 c_3) + \hat{k} (b_1 c_2 - b_2 c_1) \} \\
&= \left| \begin{array}{ccc} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ (b_2 c_3 - b_3 c_2) & (b_3 c_1 - b_1 c_3) & (b_1 c_2 - b_2 c_1) \end{array} \right|
\end{aligned}$$

$$\begin{aligned}
&= \hat{i}\{a_2(b_1c_2 - b_2c_1) - a_3(b_3c_1 - b_1c_3)\} - \hat{j}\{a_1(b_1c_2 - b_2c_1) - a_3(b_2c_3 - b_3c_2)\} \\
&+ \hat{k}\{a_1(b_3c_1 - b_1c_3) - a_2(b_2c_3 - b_3c_2)\} \\
&= \hat{i}(a_2b_1c_2 - a_2b_2c_1 - a_3b_3c_1 + a_3b_1c_3) + \hat{j}(-a_1b_1c_2 + a_1b_2c_1 + a_3b_2c_3 - a_3b_3c_2) \\
&+ \hat{k}(a_1b_3c_1 - a_1b_1c_3 - a_2b_2c_3 + a_2b_3c_2) \\
&= \hat{i}(a_1b_1c_1 - a_1b_1c_1 + a_2b_1c_2 - a_2b_2c_1 - a_3b_3c_1 + a_3b_1c_3) + \\
&\hat{j}(a_2b_2c_2 - a_2b_2c_2 - a_1b_1c_2 + a_1b_2c_1 + a_3b_2c_3 - a_3b_3c_2) \\
&+ \hat{k}(a_3b_3c_3 - a_3b_3c_3 + a_1b_3c_1 - a_1b_1c_3 - a_2b_2c_3 + a_2b_3c_2) \\
&= \hat{i}\{(a_1c_1 + a_2c_2 + a_3c_3)b_1 - (a_1b_1 + a_2b_2 + a_3b_3)c_1\} + \hat{j}\{(a_2c_2 + a_1c_1 + a_3c_3)b_2 \\
&- (a_2b_2 + a_1b_1 + a_3b_3)c_2\} + \hat{k}\{(a_3c_3 + a_1c_1 + a_2c_2)b_3 - (a_3b_3 + a_1b_1 + a_2b_2)c_3\} \\
&= \hat{i}\{(a_1c_1 + a_2c_2 + a_3c_3)b_1 - (a_1b_1 + a_2b_2 + a_3b_3)c_1\} + \hat{j}\{(a_1c_1 + a_2c_2 + a_3c_3)b_2 \\
&- (a_1b_1 + a_2b_2 + a_3b_3)c_2\} + \hat{k}\{(a_1c_1 + a_2c_2 + a_3c_3)b_3 - (a_1b_1 + a_2b_2 + a_3b_3)c_3\} \\
&= \hat{i}(a_1c_1 + a_2c_2 + a_3c_3)b_1 + \hat{j}(a_1c_1 + a_2c_2 + a_3c_3)b_2 + \hat{k}(a_1c_1 + a_2c_2 + a_3c_3)b_3 \\
&- \hat{i}(a_1b_1 + a_2b_2 + a_3b_3)c_1 - \hat{j}(a_1b_1 + a_2b_2 + a_3b_3)c_2 - \hat{k}(a_1b_1 + a_2b_2 + a_3b_3)c_3 \\
&= \hat{i}(a_1c_1 + a_2c_2 + a_3c_3)b_1 + \hat{j}(a_1c_1 + a_2c_2 + a_3c_3)b_2 + \hat{k}(a_1c_1 + a_2c_2 + a_3c_3)b_3 \\
&- \{\hat{i}(a_1b_1 + a_2b_2 + a_3b_3)c_1 + \hat{j}(a_1b_1 + a_2b_2 + a_3b_3)c_2 + \hat{k}(a_1b_1 + a_2b_2 + a_3b_3)c_3\} \\
&= (a_1c_1 + a_2c_2 + a_3c_3)(b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) - (a_1b_1 + a_2b_2 + a_3b_3)(c_1\hat{i} + c_2\hat{j} + c_3\hat{k}) \\
&= \{(a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \cdot (c_1\hat{i} + c_2\hat{j} + c_3\hat{k})\} (b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) - \{(a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \cdot \\
&(b_1\hat{i} + b_2\hat{j} + b_3\hat{k})\} (c_1\hat{i} + c_2\hat{j} + c_3\hat{k}) \\
&= (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} \text{ Proved}
\end{aligned}$$

Q# 50:

a) Prove that $\nabla \times (\nabla \times \vec{A}) = -\nabla^2 \vec{A} + \nabla(\nabla \cdot \vec{A})$

b) Prove that $\nabla \times (\phi \vec{A}) = (\nabla \phi) \times \vec{A} + \phi(\nabla \times \vec{A})$

Answer a)

Let, $\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$

$$\begin{aligned}
\therefore \nabla \times \vec{A} &= (\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}) \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\
&= \hat{i} [\frac{\partial}{\partial y} (A_3) - \frac{\partial}{\partial z} (A_2)] - \hat{j} [\frac{\partial}{\partial x} (A_3) - \frac{\partial}{\partial z} (A_1)] + \hat{k} [\frac{\partial}{\partial x} (A_2) - \frac{\partial}{\partial y} (A_1)] \\
&= \hat{i} [\frac{\partial}{\partial y} (A_3) - \frac{\partial}{\partial z} (A_2)] + \hat{j} [\frac{\partial}{\partial z} (A_1) - \frac{\partial}{\partial x} (A_3)] + \hat{k} [\frac{\partial}{\partial x} (A_2) - \frac{\partial}{\partial y} (A_1)] \\
\therefore \nabla \times (\nabla \times \vec{A}) &= (\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}) \times \{ \hat{i} [\frac{\partial}{\partial y} (A_3) - \frac{\partial}{\partial z} (A_2)] + \hat{j} [\frac{\partial}{\partial z} (A_1) - \frac{\partial}{\partial x} (A_3)] \\
&\quad + \hat{k} [\frac{\partial}{\partial x} (A_2) - \frac{\partial}{\partial y} (A_1)] \} \\
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} & \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} & \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \end{vmatrix} \\
&= \hat{i} [\frac{\partial}{\partial y} (\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y}) - \frac{\partial}{\partial z} (\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x})] - \hat{j} [\frac{\partial}{\partial x} (\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y}) \\
&\quad - \frac{\partial}{\partial z} (\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z})] + \hat{k} [\frac{\partial}{\partial x} (\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}) - \frac{\partial}{\partial y} (\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z})] \\
&= \hat{i} [\frac{\partial^2 A_2}{\partial y \partial x} - \frac{\partial^2 A_1}{\partial y^2}] - (\frac{\partial^2 A_1}{\partial z^2} - \frac{\partial^2 A_3}{\partial z \partial x}) - \hat{j} [(\frac{\partial^2 A_2}{\partial x^2} - \frac{\partial^2 A_1}{\partial x \partial y}) - (\frac{\partial^2 A_3}{\partial z \partial y} - \frac{\partial^2 A_2}{\partial z^2})] \\
&\quad + \hat{k} [(\frac{\partial^2 A_1}{\partial x \partial z} - \frac{\partial^2 A_3}{\partial x^2}) - (\frac{\partial^2 A_3}{\partial y^2} - \frac{\partial^2 A_2}{\partial y \partial z})] \\
&= \hat{i} [-\frac{\partial^2 A_1}{\partial y^2} - \frac{\partial^2 A_1}{\partial z^2}] + \hat{j} [-\frac{\partial^2 A_2}{\partial x^2} - \frac{\partial^2 A_2}{\partial z^2}] + \hat{k} [-\frac{\partial^2 A_3}{\partial x^2} - \frac{\partial^2 A_3}{\partial y^2}] + \hat{i} [\frac{\partial^2 A_2}{\partial y \partial x} \\
&\quad + \frac{\partial^2 A_3}{\partial z \partial x}] + \hat{j} [\frac{\partial^2 A_1}{\partial x \partial y} + \frac{\partial^2 A_3}{\partial z \partial y}] + \hat{k} [\frac{\partial^2 A_1}{\partial x \partial z} + \frac{\partial^2 A_2}{\partial y \partial z}]
\end{aligned}$$

$$\begin{aligned}
&= \hat{i} \left[-\frac{\delta^2 A_1}{\delta x^2} - \frac{\delta^2 A_1}{\delta y^2} - \frac{\delta^2 A_1}{\delta z^2} \right] + \hat{j} \left[-\frac{\delta^2 A_2}{\delta y^2} - \frac{\delta^2 A_2}{\delta x^2} - \frac{\delta^2 A_2}{\delta z^2} \right] \\
&+ \hat{k} \left[-\frac{\delta^2 A_3}{\delta z^2} - \frac{\delta^2 A_3}{\delta x^2} - \frac{\delta^2 A_3}{\delta y^2} \right] + \hat{i} \left[\frac{\delta^2 A_1}{\delta x^2} + \frac{\delta^2 A_2}{\delta y \delta x} + \frac{\delta^2 A_3}{\delta z \delta x} \right] + \hat{j} \left[\frac{\delta^2 A_1}{\delta x \delta y} \right. \\
&+ \left. \frac{\delta^2 A_2}{\delta y^2} + \frac{\delta^2 A_3}{\delta z \delta y} \right] + \hat{k} \left[\frac{\delta^2 A_3}{\delta z^2} + \frac{\delta^2 A_1}{\delta x \delta z} + \frac{\delta^2 A_2}{\delta y \delta z} \right] \\
&= -\left(\frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2} \right) A_1 \hat{i} - \left(\frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2} \right) A_2 \hat{j} - \left(\frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} \right. \\
&+ \left. \frac{\delta^2}{\delta z^2} \right) A_3 \hat{k} + \hat{i} \frac{\delta}{\delta x} \left[\frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right] + \hat{j} \frac{\delta}{\delta y} \left[\frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right] \\
&+ \hat{k} \frac{\delta}{\delta z} \left[\frac{\delta A_3}{\delta z} + \frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} \right] \\
&= -\left(\frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2} \right) A_1 \hat{i} - \left(\frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2} \right) A_2 \hat{j} - \left(\frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} \right. \\
&+ \left. \frac{\delta^2}{\delta z^2} \right) A_3 \hat{k} + \hat{i} \frac{\delta}{\delta x} \left[\frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right] + \hat{j} \frac{\delta}{\delta y} \left[\frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right] \\
&+ \hat{k} \frac{\delta}{\delta z} \left[\frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right] \\
&= -\left(\frac{\delta^2}{\delta x^2} + \frac{\delta^2}{\delta y^2} + \frac{\delta^2}{\delta z^2} \right) (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \left(\frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} \right. \\
&+ \left. \frac{\delta A_3}{\delta z} \right) \\
&= -\nabla^2 (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \nabla \left(\frac{\delta A_1}{\delta x} + \frac{\delta A_2}{\delta y} + \frac{\delta A_3}{\delta z} \right) \\
&= -\nabla^2 \vec{A} + \nabla \left[\left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \right] \\
&= -\nabla^2 \vec{A} + \nabla (\nabla \cdot \vec{A}) \text{ Proved}
\end{aligned}$$

Answer b)

$$\text{Let, } \vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$$

$$\phi \vec{A} = \phi (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k})$$

$$\vec{\phi A} = \phi A_1 \hat{i} + \phi A_2 \hat{j} + \phi A_3 \hat{k}$$

$$\text{L.H.S. } \vec{\nabla} \times (\phi \vec{A}) = \left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \times (\phi A_1 \hat{i} + \phi A_2 \hat{j} + \phi A_3 \hat{k})$$

$$\begin{aligned}
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ \phi A_1 & \phi A_2 & \phi A_3 \end{vmatrix} \\
&= \hat{i} \left[\frac{\delta}{\delta y} (\phi A_3) - \frac{\delta}{\delta z} (\phi A_2) \right] - \hat{j} \left[\frac{\delta}{\delta x} (\phi A_3) - \frac{\delta}{\delta z} (\phi A_1) \right] + \hat{k} \left[\frac{\delta}{\delta x} (\phi A_2) - \frac{\delta}{\delta y} (\phi A_1) \right] \\
&= \hat{i} \left[\phi \frac{\delta}{\delta y} (A_3) + A_3 \frac{\delta \phi}{\delta y} - \phi \frac{\delta}{\delta z} (A_2) - A_2 \frac{\delta \phi}{\delta z} \right] - \hat{j} \left[\phi \frac{\delta}{\delta x} (A_3) + A_3 \frac{\delta \phi}{\delta x} - \phi \frac{\delta}{\delta z} (A_1) - A_1 \frac{\delta \phi}{\delta z} \right] \\
&\quad + \hat{k} \left[\phi \frac{\delta}{\delta x} (A_2) + A_2 \frac{\delta \phi}{\delta x} - \phi \frac{\delta}{\delta y} (A_1) - A_1 \frac{\delta \phi}{\delta y} \right] \\
&= \phi \left[\hat{i} \left\{ \frac{\delta}{\delta y} (A_3) - \frac{\delta}{\delta z} (A_2) \right\} - \hat{j} \left\{ \frac{\delta}{\delta x} (A_3) - \frac{\delta}{\delta z} (A_1) \right\} + \hat{k} \left\{ \frac{\delta}{\delta x} (A_2) - \frac{\delta}{\delta y} (A_1) \right\} \right] \\
&\quad + \hat{i} \left[A_3 \frac{\delta \phi}{\delta y} - A_2 \frac{\delta \phi}{\delta z} \right] - \hat{j} \left[A_3 \frac{\delta \phi}{\delta x} - A_1 \frac{\delta \phi}{\delta z} \right] + \hat{k} \left[A_2 \frac{\delta \phi}{\delta x} - A_1 \frac{\delta \phi}{\delta y} \right] \\
&= \phi \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ A_1 & A_2 & A_3 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta \phi}{\delta x} & \frac{\delta \phi}{\delta y} & \frac{\delta \phi}{\delta z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\
&= \phi \left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left(\hat{i} \frac{\delta \phi}{\delta x} + \hat{j} \frac{\delta \phi}{\delta y} + \hat{k} \frac{\delta \phi}{\delta z} \right) \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\
&= \phi \left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left(\hat{i} \frac{\delta \phi}{\delta x} + \hat{j} \frac{\delta \phi}{\delta y} + \hat{k} \frac{\delta \phi}{\delta z} \right) \phi \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\
&= \phi (\vec{\nabla} \times \vec{A}) + (\vec{\nabla} \phi) \times \vec{A} \quad (\text{Proved})
\end{aligned}$$

Q# 51: If $r^2 = x^2 + y^2 + z^2$ then find $\frac{\partial r}{\partial x}$, $\frac{\partial r}{\partial y}$, $\frac{\partial r}{\partial z}$

We have, $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$

$$\therefore \left| \vec{r} \right| = r = \sqrt{x^2 + y^2 + z^2} = (x^2 + y^2 + z^2)^{1/2}$$

$$\therefore \left| \vec{r} \right|^2 = r^2 = \left\{ (x^2 + y^2 + z^2)^{1/2} \right\}^2 = x^2 + y^2 + z^2$$

$$r^2 = x^2 + y^2 + z^2 \text{-----(i)}$$

Differentiating (i) with respect to x partially,

$$\therefore 2r \frac{\partial r}{\partial x} = \frac{\partial}{\partial x} (x^2 + y^2 + z^2)$$

$$\therefore 2r \frac{\partial r}{\partial x} = 2x + 0 + 0$$

$$\therefore \frac{\partial r}{\partial x} = \frac{x}{r}$$

similarly

$$\frac{\partial r}{\partial y} = \frac{y}{r},$$

$$\frac{\partial r}{\partial z} = \frac{z}{r}$$

Q# 52: Show that $\vec{\nabla} \cdot \vec{r} = 3$

$$\begin{aligned} \text{Also, } \vec{\nabla} \cdot \vec{r} &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (x\hat{i} + y\hat{j} + z\hat{k}) \\ &= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3 \end{aligned}$$

$$[\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1, \hat{i} \cdot \hat{j} = 0, \hat{i} \cdot \hat{k} = 0, \hat{j} \cdot \hat{i} = 0, \hat{j} \cdot \hat{k} = 0, \hat{k} \cdot \hat{i} = 0, \hat{k} \cdot \hat{j} = 0]$$

Q# 53: Show that $\vec{r} \cdot \vec{r} = r^2$

$$\text{Also, } \vec{r} \cdot \vec{r} = (x\hat{i} + y\hat{j} + z\hat{k}) \cdot (x\hat{i} + y\hat{j} + z\hat{k})$$

$$\begin{aligned} \vec{r} \cdot \vec{r} &= x^2 + y^2 + z^2 [\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1] \\ &= r^2 [\because r^2 = x^2 + y^2 + z^2] \end{aligned}$$

Similarly,

$$\vec{A} \cdot \vec{A} = A^2$$

$$\vec{\nabla} \cdot \vec{\nabla} = \nabla^2$$

Q# 54: Show that, $\vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}$

Proof: L.H.S = $\vec{\nabla}$

$$\begin{aligned} &= \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \\ &= \hat{i} \frac{\partial}{\partial x} \frac{\partial r}{\partial r} + \hat{j} \frac{\partial}{\partial y} \frac{\partial r}{\partial r} + \hat{k} \frac{\partial}{\partial z} \frac{\partial r}{\partial r} \\ &= \hat{i} \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \hat{j} \frac{\partial r}{\partial y} \frac{\partial}{\partial r} + \hat{k} \frac{\partial r}{\partial z} \frac{\partial}{\partial r} \\ &= \left(\hat{i} \frac{\partial r}{\partial x} + \hat{j} \frac{\partial r}{\partial y} + \hat{k} \frac{\partial r}{\partial z} \right) \frac{\partial}{\partial r} \\ &= \left(\hat{i} \frac{x}{r} + \hat{j} \frac{y}{r} + \hat{k} \frac{z}{r} \right) \frac{\partial}{\partial r} [\because \frac{\partial r}{\partial x} = \frac{x}{r}; \frac{\partial r}{\partial y} = \frac{y}{r}; \frac{\partial r}{\partial z} = \frac{z}{r}] \end{aligned}$$

$$= \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r} \frac{\partial}{\partial r}$$

$$= \frac{\vec{r}}{r} \frac{\partial}{\partial r}$$

Q# 55: Show that $\vec{\nabla} \cdot (\phi \vec{A}) = \phi(\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}$

Solution

Let $\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$ is a vector and ϕ is a function of a variable or variables

$$\begin{aligned} \text{L.H.S } \vec{\nabla} \cdot (\phi \vec{A}) &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (\phi \vec{A}) \\ &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \phi (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\ &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (\phi A_1 \hat{i} + \phi A_2 \hat{j} + \phi A_3 \hat{k}) \\ &= \frac{\partial}{\partial x} (\phi A_1) + \frac{\partial}{\partial y} (\phi A_2) + \frac{\partial}{\partial z} (\phi A_3) \quad [\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1] \\ &= \phi \frac{\partial}{\partial x} (A_1) + A_1 \frac{\partial \phi}{\partial x} + \phi \frac{\partial}{\partial y} (A_2) + A_2 \frac{\partial \phi}{\partial y} + \phi \frac{\partial}{\partial z} (A_3) + A_3 \frac{\partial \phi}{\partial z} \\ &\quad [\because \frac{d}{dx} (uv) = u \frac{d}{dx} v + v \frac{d}{dx} u] \\ &= A_1 \frac{\partial \phi}{\partial x} + A_2 \frac{\partial \phi}{\partial y} + A_3 \frac{\partial \phi}{\partial z} + \phi \frac{\partial}{\partial x} (A_1) + \phi \frac{\partial}{\partial y} (A_2) + \phi \frac{\partial}{\partial z} (A_3) \\ &= A_1 \frac{\partial \phi}{\partial x} + A_2 \frac{\partial \phi}{\partial y} + A_3 \frac{\partial \phi}{\partial z} + \phi \left\{ \frac{\partial}{\partial x} (A_1) + \frac{\partial}{\partial y} (A_2) + \frac{\partial}{\partial z} (A_3) \right\} \\ &= A_1 \frac{\partial \phi}{\partial x} + A_2 \frac{\partial \phi}{\partial y} + A_3 \frac{\partial \phi}{\partial z} + \phi \left\{ \frac{\partial}{\partial x} (A_1) + \frac{\partial}{\partial y} (A_2) + \frac{\partial}{\partial z} (A_3) \right\} \\ &= \frac{\partial \phi}{\partial x} A_1 + \frac{\partial \phi}{\partial y} A_2 + \frac{\partial \phi}{\partial z} A_3 + \phi \left\{ \frac{\partial}{\partial x} (A_1) + \frac{\partial}{\partial y} (A_2) + \frac{\partial}{\partial z} (A_3) \right\} \\ &= \phi \left\{ \frac{\partial}{\partial x} (A_1) + \frac{\partial}{\partial y} (A_2) + \frac{\partial}{\partial z} (A_3) \right\} + \frac{\partial \phi}{\partial x} A_1 + \frac{\partial \phi}{\partial y} A_2 + \frac{\partial \phi}{\partial z} A_3 \\ &= \phi \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\ &\quad [\because \hat{i} \cdot \hat{i} = 1; \hat{j} \cdot \hat{j} = 1; \hat{k} \cdot \hat{k} = 1] \\ &= \phi \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) + \left(\frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k} \right) \cdot \phi (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \end{aligned}$$

$$= \phi(\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}$$

$$[\because \vec{\nabla} = \frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k}]$$

$$\therefore \vec{\nabla} \cdot (\phi \vec{A}) = \phi(\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}$$

Q# 56: Show that $\nabla^2(\ln r) = \frac{1}{r^2}$

L.H.S = $\nabla^2(\ln r)$

$$= \vec{\nabla} \cdot \vec{\nabla}(\ln r)$$

$$[\because \vec{\nabla} \cdot \vec{\nabla} = \nabla^2]$$

$$= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} \ln r \right]$$

$$[\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} (\ln r) \right]$$

$$= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{1}{r} \right]$$

$$[\because \frac{\partial}{\partial r} (\ln r) = \frac{1}{r}]$$

$$= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r^2} \right]$$

$$= \vec{\nabla} \cdot \left[\frac{1}{r^2} \vec{r} \right]$$

$$= \frac{1}{r^2} [\vec{\nabla} \cdot \vec{r}] + \left[\vec{\nabla} \left(\frac{1}{r^2} \right) \right] \cdot \vec{r}$$

$$[\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi(\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}]$$

$$= \frac{3}{r^2} + \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} \left(\frac{1}{r^2} \right) \right] \cdot \vec{r}$$

$$[\because \vec{\nabla} \cdot \vec{r} = 3 \text{ \& } \because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$= \frac{3}{r^2} + \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-2}) \right] \cdot \vec{r}$$

$$= \frac{3}{r^2} + \frac{\vec{r}}{r} (-2r^{-2-1}) \cdot \vec{r}$$

$$= \frac{3}{r^2} + \frac{\vec{r}}{r} \left(-\frac{2}{r^3} \right) \cdot \vec{r}$$

$$= \frac{3}{r^2} - \frac{2}{r^4} (\vec{r} \cdot \vec{r})$$

$$= \frac{3}{r^2} - \frac{2}{r^4} \times r^2$$

$$[\because \vec{r} \cdot \vec{r} = r^2]$$

$$= \frac{3}{r^2} - \frac{2}{r^2}$$

$$= \frac{1}{r^2}.$$

Q# 57: Find the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r}$

Answer: Let $\phi = \frac{1}{r}$

Therefore, the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r} = \nabla \phi \cdot \hat{r}$

We can write, $\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{\vec{r}}{r}$

Here, $\phi = \frac{1}{r}$

$$\Rightarrow \vec{\nabla} \phi = \vec{\nabla} \frac{1}{r}$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} \frac{\partial}{\partial r} \frac{1}{r} \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-1})$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} (-1)(r^{-1-1})$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} (-1)(r^{-2})$$

$$\Rightarrow \vec{\nabla} \phi = -\frac{\vec{r}}{r} \frac{1}{r^2}$$

$$\Rightarrow \vec{\nabla} \phi = -\frac{\vec{r}}{r^3}$$

Therefore, the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r}$

$$= \vec{\nabla} \phi \cdot \hat{r}$$

$$= -\frac{\vec{r}}{r^3} \cdot \hat{r}$$

$$= -\frac{\vec{r}}{r^3} \cdot \frac{\vec{r}}{r}$$

$$= -\frac{1}{r^4} (\vec{r} \cdot \vec{r})$$

$$= -\frac{1}{r^4} r^2$$

$$[\vec{\nabla} \phi = -\frac{\vec{r}}{r^3}]$$

$$[\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{\vec{r}}{r}]$$

$$[\because \vec{r} \cdot \vec{r} = r^2]$$

$$= -\frac{1}{r^2}$$

Q# 58: Show that $\nabla^2 \mathbf{r}^n = n(n+1)\mathbf{r}^{n-2}$

$$\begin{aligned}
\text{L.H.S} &= \nabla^2 \mathbf{r}^n = \vec{\nabla} \cdot \vec{\nabla}(\mathbf{r}^n) & [\because \vec{\nabla} \cdot \vec{\nabla} = \nabla^2] \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} \mathbf{r}^n \right] & [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} n \mathbf{r}^{n-1} \right] \\
&= \vec{\nabla} \cdot \left[n \vec{r} r^{n-1} r^{-1} \right] \\
&= \vec{\nabla} \cdot \left[n \vec{r} r^{n-1-1} \right] \\
&= \vec{\nabla} \cdot \left[n \vec{r} r^{n-2} \right] \\
&= n \left[\vec{\nabla} \cdot (\vec{r} r^{n-2}) \right] \\
&= n \left[\vec{\nabla} \cdot (r^{n-2} \vec{r}) \right] \\
&= n \left[r^{n-2} (\vec{\nabla} \cdot \vec{r}) + \vec{\nabla}(r^{n-2}) \cdot \vec{r} \right] & [\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}] \\
&= n \left[3r^{n-2} + \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{n-2}) \cdot \vec{r} \right] & [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r} \text{ \& } [\vec{\nabla} \cdot \vec{r} = 3]] \\
&= n \left[3r^{n-2} + \frac{1}{r} \frac{\partial}{\partial r} (r^{n-2}) (\vec{r} \cdot \vec{r}) \right] \\
&= n \left[3r^{n-2} + \frac{1}{r} (n-2)(r^{n-2-1}) (\vec{r} \cdot \vec{r}) \right] \\
&= n \left[3r^{n-2} + \frac{1}{r} (n-2) r^{n-3} r^2 \right] & [\because \vec{r} \cdot \vec{r} = r^2] \\
&= n \left[3r^{n-2} + (n-2) r^{n-3} r \right] \\
&= n \left[3r^{n-2} + (n-2) r^{n-3+1} \right] \\
&= n \left[3r^{n-2} + (n-2) r^{n-2} \right] \\
&= n[3 + n - 2] r^{n-2} \\
&= n(n+1) r^{n-2}
\end{aligned}$$

Q# 59: Show that $\vec{\nabla} \cdot (r^3 \vec{r}) = 6r^3$

$$\begin{aligned}
\text{L.H.S} &= \vec{\nabla} \cdot (r^3 \vec{r}) \\
&= r^3 (\vec{\nabla} \cdot \vec{r}) + \vec{\nabla}(r^3) \cdot \vec{r} & [\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}] \\
&= 3r^3 + \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} r^3 \right] \cdot \vec{r} & [\because \vec{\nabla} \cdot \vec{r} = 3 \text{ \& } \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]
\end{aligned}$$

$$\begin{aligned}
&= 3\mathbf{r}^3 + \left[\frac{\vec{\mathbf{r}}}{\mathbf{r}} 3\mathbf{r}^{3-1} \right] \cdot \vec{\mathbf{r}} \\
&= 3\mathbf{r}^3 + \frac{1}{\mathbf{r}} 3\mathbf{r}^2 (\vec{\mathbf{r}} \cdot \vec{\mathbf{r}}) \\
&= 3\mathbf{r}^3 + 3\mathbf{r}(\mathbf{r}^2) \quad [\because \vec{\mathbf{r}} \cdot \vec{\mathbf{r}} = \mathbf{r}^2] \\
&= 3\mathbf{r}^3 + 3\mathbf{r}^3 \\
&= 6\mathbf{r}^3
\end{aligned}$$

Q# 60: Show that $\vec{\nabla} \cdot \left[\mathbf{r} \vec{\nabla} \left(\frac{1}{\mathbf{r}^3} \right) \right] = \frac{3}{\mathbf{r}^4}$.

$$\begin{aligned}
\text{L.H.S} &= \vec{\nabla} \cdot \left[\mathbf{r} \vec{\nabla} \left(\frac{1}{\mathbf{r}^3} \right) \right] \\
&= \vec{\nabla} \cdot \left[\mathbf{r} \left\{ \frac{\vec{\mathbf{r}}}{\mathbf{r}} \frac{\partial}{\partial \mathbf{r}} \left(\frac{1}{\mathbf{r}^3} \right) \right\} \right] \quad [\because \vec{\nabla} = \frac{\vec{\mathbf{r}}}{\mathbf{r}} \frac{\partial}{\partial \mathbf{r}}] \\
&= \vec{\nabla} \cdot \left[\mathbf{r} \left\{ \frac{\vec{\mathbf{r}}}{\mathbf{r}} \frac{\partial}{\partial \mathbf{r}} \mathbf{r}^{-3} \right\} \right] \\
&= \vec{\nabla} \cdot \left[\vec{\mathbf{r}} \frac{\partial}{\partial \mathbf{r}} \mathbf{r}^{-3} \right] \\
&= \vec{\nabla} \cdot [\vec{\mathbf{r}}(-3)\mathbf{r}^{-3-1}] \\
&= \vec{\nabla} \cdot [\vec{\mathbf{r}}(-3)\mathbf{r}^{-4}] \\
&= \vec{\nabla} \cdot [-3\mathbf{r}^{-4}\vec{\mathbf{r}}] \\
&= -3\mathbf{r}^{-4}(\vec{\nabla} \cdot \vec{\mathbf{r}}) + \left\{ \vec{\nabla}(-3\mathbf{r}^{-4}) \right\} \cdot \vec{\mathbf{r}} \quad [\because \vec{\nabla} \cdot (\phi \vec{\mathbf{A}}) = \phi(\vec{\nabla} \cdot \vec{\mathbf{A}}) + (\vec{\nabla} \phi) \cdot \vec{\mathbf{A}}] \\
&= -3\mathbf{r}^{-4}(\vec{\nabla} \cdot \vec{\mathbf{r}}) - 3 \left\{ \vec{\nabla}(\mathbf{r}^{-4}) \right\} \cdot \vec{\mathbf{r}} \\
&= -9\mathbf{r}^{-4} - 3 \left\{ \frac{\vec{\mathbf{r}}}{\mathbf{r}} \frac{\partial}{\partial \mathbf{r}} \mathbf{r}^{-4} \right\} \cdot \vec{\mathbf{r}} \quad [\because \vec{\nabla} \cdot \vec{\mathbf{r}} = 3] \text{ \& } [\because \vec{\nabla} = \frac{\vec{\mathbf{r}}}{\mathbf{r}} \frac{\partial}{\partial \mathbf{r}}] \\
&= -9\mathbf{r}^{-4} - 3 \left\{ \frac{\vec{\mathbf{r}}}{\mathbf{r}} \frac{\partial}{\partial \mathbf{r}} \mathbf{r}^{-4} \right\} \cdot \vec{\mathbf{r}} \\
&= -9\mathbf{r}^{-4} - 3 \left\{ \frac{\vec{\mathbf{r}}}{\mathbf{r}} (-4)\mathbf{r}^{-4-1} \right\} \cdot \vec{\mathbf{r}} \\
&= -9\mathbf{r}^{-4} - 3 \left\{ (-4)\mathbf{r}^{-4-1} \times \frac{1}{\mathbf{r}} \right\} (\vec{\mathbf{r}} \cdot \vec{\mathbf{r}}) \\
&= -\frac{9}{\mathbf{r}^4} - 3(-4)\mathbf{r}^{-5}\mathbf{r}^{-1}(\vec{\mathbf{r}} \cdot \vec{\mathbf{r}}) \\
&= -\frac{9}{\mathbf{r}^4} - 3(-4)\mathbf{r}^{-6}(\vec{\mathbf{r}} \cdot \vec{\mathbf{r}})
\end{aligned}$$

$$\begin{aligned}
&= -\frac{9}{r^4} + \frac{12}{r^6} \cdot r^2 \\
&= -\frac{9}{r^4} + \frac{12}{r^4} \\
&= \frac{3}{r^4}.
\end{aligned}$$

$$[\because \vec{r} \cdot \vec{r} = r^2]$$

Q# 61: Show that $\nabla^2 \left[\vec{\nabla} \cdot \left(\frac{\vec{r}}{r^2} \right) \right] = \frac{2}{r^4}.$

$$\begin{aligned}
\text{L.H.S} &= \nabla^2 \left[\vec{\nabla} \cdot \left(\frac{\vec{r}}{r^2} \right) \right] \\
&= \nabla^2 \left[\vec{\nabla} \cdot \left(\frac{1}{r^2} \vec{r} \right) \right] \\
&= \nabla^2 \left[\frac{1}{r^2} (\vec{\nabla} \cdot \vec{r}) + \left\{ \vec{\nabla} \cdot \left(\frac{1}{r^2} \right) \right\} \cdot \vec{r} \right] \\
&= \nabla^2 \left[\frac{3}{r^2} + \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} \left(\frac{1}{r^2} \right) \right\} \cdot \vec{r} \right] \\
&= \nabla^2 \left[\frac{3}{r^2} + \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-2}) \right\} \cdot \vec{r} \right] \\
&= \nabla^2 \left[\frac{3}{r^2} + \left\{ \frac{\vec{r}}{r} (-2)(r^{-2-1}) \right\} \cdot \vec{r} \right] \\
&= \nabla^2 \left[\frac{3}{r^2} + \left\{ \frac{\vec{r}}{r} (-2)(r^{-3}) \right\} \cdot \vec{r} \right] \\
&= \nabla^2 \left[\frac{3}{r^2} + \left\{ \vec{r} \cdot (-2)(r^{-3} \cdot r^{-1}) \right\} \cdot \vec{r} \right] \\
&= \nabla^2 \left[\frac{3}{r^2} + \left\{ \vec{r} \cdot (-2)(r^{-4}) \right\} \cdot \vec{r} \right] \\
&= \nabla^2 \left[\frac{3}{r^2} + \frac{-2}{r^4} (\vec{r} \cdot \vec{r}) \right] \\
&= \nabla^2 \left[\frac{3}{r^2} + \frac{-2}{r^4} r^2 \right] \\
&= \nabla^2 \left[\frac{3}{r^2} - \frac{2}{r^2} \right] \\
&= \nabla^2 \left(\frac{1}{r^2} \right)
\end{aligned}$$

$$[\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}]$$

$$[\because \vec{\nabla} \cdot \vec{r} = 3] \& [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$[\because \vec{r} \cdot \vec{r} = r^2]$$

$$\begin{aligned}
&= \vec{\nabla} \cdot \vec{\nabla} \left(\frac{1}{r^2} \right) \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} \left(\frac{1}{r^2} \right) \right] \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-2}) \right] \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} (-2)(r^{-2-1}) \right] \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} (-2)(r^{-3}) \right] \\
&= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \left(-\frac{2}{r^3} \right) \right] \\
&= \vec{\nabla} \cdot \left[\vec{r} \left(-\frac{2}{r^4} \right) \right] \\
&= \vec{\nabla} \cdot \left[\left(-\frac{2}{r^4} \right) \vec{r} \right] \\
&= -\frac{2}{r^4} (\vec{\nabla} \cdot \vec{r}) + \left\{ \vec{\nabla} \left(\frac{-2}{r^4} \right) \right\} \cdot \vec{r} \\
&= -\frac{2}{r^4} (\vec{\nabla} \cdot \vec{r}) - 2 \left\{ \vec{\nabla} (r^{-4}) \right\} \cdot \vec{r} \\
&= -\frac{6}{r^4} - 2 \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-4}) \right\} \cdot \vec{r} \\
&= -\frac{6}{r^4} - 2 \left\{ \frac{\vec{r}}{r} (-4)(r^{-4-1}) \right\} \cdot \vec{r} \\
&= -\frac{6}{r^4} - 2 \left\{ \frac{\vec{r}}{r} (-4)(r^{-5}) \right\} \cdot \vec{r} \\
&= -\frac{6}{r^4} - 2 \left\{ \vec{r} (-4)(r^{-5} r^{-1}) \right\} \cdot \vec{r} \\
&= -\frac{6}{r^4} + 8 \left\{ \vec{r} (r^{-6}) \right\} \cdot \vec{r} \\
&= -\frac{6}{r^4} + 8 \frac{1}{r^6} (\vec{r} \cdot \vec{r}) \\
&= -\frac{6}{r^4} + 8 \frac{1}{r^6} r^2 \\
&= -\frac{6}{r^4} + \frac{8}{r^4}
\end{aligned}$$

$$[\because \vec{\nabla} \cdot \vec{\nabla} = \nabla^2]$$

$$[\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$[\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}]$$

$$[\because \vec{\nabla} \cdot \vec{r} = 3] \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$[\because \vec{r} \cdot \vec{r} = r^2]$$

$$= \frac{2}{r^4}.$$

Q# 62: Show that $\text{grad div } \vec{A} = -2\vec{r}^{-3}\vec{r}$; Where, $\vec{A} = \frac{\vec{r}}{r}$

$$\begin{aligned} \text{Answer: } \text{grad div } \vec{A} &= \text{grad}(\vec{\nabla} \cdot \vec{A}) \\ &= \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) \end{aligned}$$

Now, **L.H.S** = $\text{grad div } \vec{A}$

$$\begin{aligned} &= \text{grad}(\vec{\nabla} \cdot \vec{A}) \\ &= \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) \\ &= \vec{\nabla}\left(\vec{\nabla} \cdot \frac{\vec{r}}{r}\right) \quad [\text{Given, } \vec{A} = \frac{\vec{r}}{r}] \\ &= \vec{\nabla}\left[\vec{\nabla} \cdot \left(\frac{1}{r}\vec{r}\right)\right] \\ &= \vec{\nabla}\left[\frac{1}{r}(\vec{\nabla} \cdot \vec{r}) + \left\{\vec{\nabla}\left(\frac{1}{r}\right)\right\} \cdot \vec{r}\right] \quad [\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi(\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}] \\ &= \vec{\nabla}\left[\frac{3}{r} + \left\{\frac{\vec{r}}{r} \frac{\partial}{\partial r}(r^{-1})\right\} \cdot \vec{r}\right] \quad [\because \vec{\nabla} \cdot \vec{r} = 3] \text{ \& } [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\ &= \vec{\nabla}\left[\frac{3}{r} + \left\{\frac{\vec{r}}{r}(-1)(r^{-1-1})\right\} \cdot \vec{r}\right] \\ &= \vec{\nabla}\left[\frac{3}{r} + \left\{\frac{\vec{r}}{r}(-1)(r^{-2})\right\} \cdot \vec{r}\right] \\ &= \vec{\nabla}\left[\frac{3}{r} + \left\{\vec{r}(-1)(r^{-2}r^{-1})\right\} \cdot \vec{r}\right] \\ &= \vec{\nabla}\left[\frac{3}{r} + \left\{\vec{r}(-1)(r^{-3})\right\} \cdot \vec{r}\right] \\ &= \vec{\nabla}\left[\frac{3}{r} + \left\{(-1)(r^{-3})\right\}(\vec{r} \cdot \vec{r})\right] \\ &= \vec{\nabla}\left[\frac{3}{r} - \frac{1}{r^3}r^2\right] \quad [\because \vec{r} \cdot \vec{r} = r^2] \\ &= \vec{\nabla}\left[\frac{3}{r} - \frac{1}{r}\right] \\ &= \vec{\nabla}\left(\frac{2}{r}\right) \end{aligned}$$

$$\begin{aligned}
&= 2 \frac{\vec{r}}{r} \frac{\partial}{\partial r} \left(\frac{1}{r} \right) & [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
&= 2 \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-1}) \\
&= 2 \frac{\vec{r}}{r} (-1) r^{-1-1} \\
&= 2 \frac{\vec{r}}{r} (-1) r^{-2} \\
&= 2 \vec{r} (-1) r^{-2} r^{-1} \\
&= 2 \vec{r} (-1) r^{-3} \\
&= -2 r^{-3} \vec{r}
\end{aligned}$$

Q# 63: i. Show that $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$

L.H.S. $= \nabla^2 f(r)$

$$\begin{aligned}
&= \vec{\nabla} \cdot \vec{\nabla} f(r) & [\because \vec{\nabla} \cdot \vec{\nabla} = \nabla^2] \\
&= \vec{\nabla} \cdot \left(\frac{\vec{r}}{r} \frac{\partial}{\partial r} f(r) \right) & [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
&= \vec{\nabla} \cdot \left(\frac{\vec{r}}{r} f'(r) \right) & [\because \frac{\partial}{\partial r} f(r) = f'(r)] \\
&= \vec{\nabla} \cdot \left[\frac{f'(r)}{r} \vec{r} \right] \\
&= \frac{f'(r)}{r} (\vec{\nabla} \cdot \vec{r}) + \left\{ \vec{\nabla} \left(\frac{f'(r)}{r} \right) \right\} \cdot \vec{r} & [\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}] \\
&= 3 \frac{f'(r)}{r} + \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} \left\{ \frac{f'(r)}{r} \right\} \right] \cdot \vec{r} & [\because \vec{\nabla} \cdot \vec{r} = 3] \\
&= 3 \frac{f'(r)}{r} + \left[\frac{1}{r} \left\{ \frac{r f''(r) - f'(r) \cdot 1}{r^2} \right\} \right] (\vec{r} \cdot \vec{r}) & [\because \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}] \\
&= 3 \frac{f'(r)}{r} + \left[\left\{ \frac{r f''(r) - f'(r) \cdot 1}{r^3} \right\} \right] (\vec{r} \cdot \vec{r}) \\
&= 3 \frac{f'(r)}{r} + \left[\left\{ \frac{r f''(r) - f'(r) \cdot 1}{r^3} \right\} \right] r^2 & [\because \vec{r} \cdot \vec{r} = r^2] \\
&= 3 \frac{f'(r)}{r} + \left[\left\{ \frac{r f''(r) - f'(r) \cdot 1}{r} \right\} \right] \\
&= 3 \frac{f'(r)}{r} + f''(r) - \frac{f'(r)}{r}
\end{aligned}$$

$$= f''(r) + \frac{2}{r} f'(r)$$

ii. If $\nabla^2 f(r) = 0$

$$f''(r) + \frac{2}{r} f'(r) = 0 \quad [\because \nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)]$$

$$\frac{\partial^2}{\partial r^2} (f(r)) + \frac{2}{r} \frac{\partial}{\partial r} (f(r)) = 0 \quad [\because \frac{\partial}{\partial r} (f(r)) = f'(r)] \text{ \& } [\because \frac{\partial^2}{\partial r^2} (f(r)) = f''(r)]$$

$$\frac{\partial}{\partial r} \left\{ \frac{\partial}{\partial r} (f(r)) \right\} + \frac{2}{r} \frac{\partial}{\partial r} (f(r)) = 0$$

$$\frac{\partial}{\partial r} (p) + \frac{2}{r} p = 0, \quad [\because p = \frac{\partial}{\partial r} (f(r)) = f'(r)]$$

$$\frac{\partial p}{\partial r} \times \frac{\partial r}{p} + \frac{2p}{r} \times \frac{\partial r}{p} = 0 \quad [\text{Multiplying by } \frac{\partial r}{p} \text{ on both sides}]$$

$$\frac{\partial p}{p} + \frac{2 \partial r}{r} = 0$$

$$\int \frac{\partial p}{p} + \int \frac{2 \partial r}{r} = \int 0 \quad [\text{Integrating}]$$

$$\ln p + 2 \ln r = \ln A$$

$$\ln p + \ln r^2 = \ln A$$

$$\ln pr^2 = \ln A \quad [\because \ln ab = \ln a + \ln b]$$

$$pr^2 = A$$

$$p = Ar^{-2}$$

$$\frac{\partial f(r)}{\partial r} = Ar^{-2}$$

$$\int \frac{\partial f(r)}{\partial r} \partial r = \int Ar^{-2} \partial r$$

$$\int \frac{\partial}{\partial r} \{f(r)\} \partial r = \int Ar^{-2} \partial r$$

Integrating

$$f(r) = A \frac{r^{-2+1}}{-2+1} + B$$

$$f(r) = A \frac{r^{-1}}{-1} + B$$

$$f(r) = -A \frac{1}{r} + B$$

$$f(r) = \frac{-A}{r} + B$$

$$\mathbf{f}(\mathbf{r}) = \mathbf{B} + \frac{\mathbf{c}}{r}, \quad \text{Where, } \mathbf{c} = -\mathbf{A}$$

Q# 64: Show that $\nabla^2 \left(\frac{1}{r} \right) = 0$

$$\begin{aligned}
 \text{L.H.S} &= \nabla^2 \left(\frac{1}{r} \right) \\
 &= \vec{\nabla} \cdot \vec{\nabla} \left(\frac{1}{r} \right) & [\because \vec{\nabla} \cdot \vec{\nabla} = \nabla^2] \\
 &= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} \left(\frac{1}{r} \right) \right] & [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
 &= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-1}) \right] \\
 &= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} (-1) r^{-1-1} \right] \\
 &= \vec{\nabla} \cdot \left[\frac{\vec{r}}{r} (-1) r^{-2} \right] \\
 &= \vec{\nabla} \cdot \left[\vec{r} (-1) r^{-2} r^{-1} \right] \\
 &= \vec{\nabla} \cdot \left[\vec{r} (-1) r^{-3} \right] \\
 &= \vec{\nabla} \cdot \left[-r^{-3} \vec{r} \right] \\
 &= \vec{\nabla} \cdot \left[-\frac{\vec{r}}{r^3} \right] \\
 &= \vec{\nabla} \cdot \left[-\vec{r} r^{-3} \right] \\
 &= \vec{\nabla} \cdot \left[-r^{-3} \vec{r} \right] \\
 &= -r^{-3} (\vec{\nabla} \cdot \vec{r}) - (\vec{\nabla} r^{-3}) \cdot \vec{r} & [\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}] \\
 &= -3r^{-3} - \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} (r^{-3}) \right\} \cdot \vec{r} & [\because \vec{\nabla} \cdot \vec{r} = 3] \text{ \& } [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
 &= -\frac{3}{r^3} - \left\{ \frac{1}{r} (-3) r^{-3-1} \right\} (\vec{r} \cdot \vec{r}) \\
 &= -\frac{3}{r^3} - \left\{ \frac{1}{r} (-3) r^{-4} \right\} (\vec{r} \cdot \vec{r}) \\
 &= -\frac{3}{r^3} - \left\{ (-3) r^{-4} r^{-1} \right\} (\vec{r} \cdot \vec{r}) \\
 &= -\frac{3}{r^3} - \left\{ (-3) r^{-4-1} \right\} (\vec{r} \cdot \vec{r})
 \end{aligned}$$

$$\begin{aligned}
&= -\frac{3}{r^3} - \{(-3)r^{-5}\}(\vec{r} \cdot \vec{r}) \\
&= -\frac{3}{r^3} - \left\{\frac{-3}{r^5}\right\}(\vec{r} \cdot \vec{r}) \\
&= -\frac{3}{r^3} - \left\{\frac{-3}{r^5}\right\}r^2 \quad [\because \vec{r} \cdot \vec{r} = r^2] \\
&= -\frac{3}{r^3} + \frac{3}{r^3} = 0
\end{aligned}$$

Q# 65: Find $\vec{\nabla} \phi$ if (a) $\phi = \ln \left| \vec{r} \right|$ (b) $\phi = \frac{1}{\left| \vec{r} \right|}$

Answer:

a) Given $\phi = \ln \left| \vec{r} \right|$

$$\vec{\nabla} \phi = \vec{\nabla} \ln \left| \vec{r} \right|$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} \frac{\partial}{\partial r} \ln \left| \vec{r} \right| \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} \frac{1}{r}$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r^2} \text{ Answer}$$

b) Given, $\phi = \frac{1}{\left| \vec{r} \right|}$

$$\vec{\nabla} \phi = \vec{\nabla} \frac{1}{\left| \vec{r} \right|}$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} \frac{\partial}{\partial r} \frac{1}{\left| \vec{r} \right|} \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} \frac{\partial}{\partial r} \left| \vec{r} \right|^{-1}$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} (-1) \left| \vec{r} \right|^{-1-1} \quad [\because \frac{d}{dx}(x^n) = nx^{n-1}]$$

$$\Rightarrow \vec{\nabla} \phi = \frac{\vec{r}}{r} (-1) \left| \vec{r} \right|^{-2}$$

$$\Rightarrow \vec{\nabla} \phi = -\frac{\vec{r}}{r} \left| \vec{r} \right|^{-2}$$

$$\Rightarrow \vec{\nabla} \phi = -\frac{\vec{r}}{r} \frac{1}{\left| \vec{r} \right|^2}$$

$$\Rightarrow \vec{\nabla} \phi = -\frac{\vec{r}}{r} \frac{1}{r^2}$$

$$\Rightarrow \vec{\nabla} \phi = -\frac{\vec{r}}{r^3} \text{ Answer}$$

c) Prove that i. $\vec{\nabla} \cdot \left[\frac{\vec{f}(\vec{r})}{r} \right] = \vec{f}'(\vec{r}) + \frac{2}{r} \vec{f}(\vec{r})$

ii. $\vec{\nabla} \cdot \left[\frac{\vec{f}(\vec{r})}{r} \right] = \frac{1}{r^2} \frac{d}{dr} [r^2 \vec{f}(\vec{r})]$

Answer: i. $\vec{\nabla} \cdot \left[\frac{\vec{f}(\vec{r})}{r} \right]$

$$= \frac{\vec{f}(\vec{r})}{r} (\vec{\nabla} \cdot \vec{r}) + \left\{ \vec{\nabla} \frac{\vec{f}(\vec{r})}{r} \right\} \cdot \vec{r} \quad [\because \vec{\nabla} \cdot (\phi \vec{A}) = \phi (\vec{\nabla} \cdot \vec{A}) + (\vec{\nabla} \phi) \cdot \vec{A}]$$

$$= \frac{3 \cdot \vec{f}(\vec{r})}{r} + \left\{ \frac{\vec{r}}{r} \frac{\partial}{\partial r} \left(\frac{\vec{f}(\vec{r})}{r} \right) \right\} \cdot \vec{r} \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}]$$

$$= \frac{3 \cdot \vec{f}(\vec{r})}{r} + \left\{ \frac{\vec{r}}{r} \left(\frac{r \vec{f}'(\vec{r}) - \vec{f}(\vec{r}) \cdot 1}{r^2} \right) \right\} \cdot \vec{r}$$

$$= \frac{3 \cdot \vec{f}(\vec{r})}{r} + \left\{ \left(\frac{r \vec{f}'(\vec{r}) - \vec{f}(\vec{r}) \cdot 1}{r^3} \right) \right\} (\vec{r} \cdot \vec{r})$$

$$= \frac{3 \cdot \vec{f}(\vec{r})}{r} + \left\{ \left(\frac{r \vec{f}'(\vec{r}) - \vec{f}(\vec{r}) \cdot 1}{r^3} \right) \right\} r^2 \quad [\because \vec{r} \cdot \vec{r} = r^2]$$

$$= \frac{3 \cdot \vec{f}(\vec{r})}{r} + \left\{ \left(\frac{r \vec{f}'(\vec{r}) - \vec{f}(\vec{r}) \cdot 1}{r} \right) \right\}$$

$$\begin{aligned}
&= \frac{3.f(r) + r f'(r) - f(r).1}{r} \\
&= \frac{3.f(r) + r f'(r) - f(r)}{r} \\
&= \frac{2f(r) + r f'(r)}{r} \\
&= \frac{2f(r)}{r} + f'(r) \\
&= f'(r) + \frac{2f(r)}{r} \\
&= f'(r) + \frac{2f(r)}{r} \quad (\text{Proved}) \\
&= \frac{1}{r^2} [r^2 f'(r) + 2r f(r)] \\
&= \frac{1}{r^2} \frac{d}{dr} [r^2 f(r)] \quad (\text{Proved})
\end{aligned}$$

Q# 66: Show that: $\vec{\nabla} r^n = n r^{n-2} \vec{r}$

Answer: L.H.S.

$$\begin{aligned}
&\vec{\nabla} r^n \\
&= \frac{\vec{r}}{r} \frac{\partial}{\partial r} r^n \quad [\because \vec{\nabla} = \frac{\vec{r}}{r} \frac{\partial}{\partial r}] \\
&= \frac{\vec{r}}{r} n r^{n-1} \\
&= \vec{r}. n r^{n-1} . r^{-1} \\
&= \vec{r}. n r^{n-2} \\
&= n r^{n-2} \vec{r} \quad (\text{Proved})
\end{aligned}$$

Q# 67: Evaluate $\vec{\nabla} . (\vec{A} \times \vec{r})$ if $\vec{\nabla} \times \vec{A} = \vec{0}$

Answer:

$$\begin{aligned}
\vec{\nabla} . (\vec{A} \times \vec{r}) &= \vec{r} . (\vec{\nabla} \times \vec{A}) - \vec{A} . (\vec{\nabla} \times \vec{r}) \quad [\because \vec{\nabla} . (\vec{A} \times \vec{B}) = \vec{B} . (\vec{\nabla} \times \vec{A}) - \vec{A} . (\vec{\nabla} \times \vec{B})] \\
&= \vec{r} . \vec{0} - \vec{A} . (\vec{\nabla} \times \vec{r}) \\
&= \vec{r} . \vec{0} - \vec{A} . \left\{ \vec{\nabla} \times (x \hat{i} + y \hat{j} + z \hat{k}) \right\}
\end{aligned}$$

$$\begin{aligned}
&= \vec{0} - \vec{A} \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ x & y & z \end{vmatrix} \\
&= \vec{A} \cdot \hat{i} \left[\frac{\delta}{\delta y}(z) - \frac{\delta}{\delta z}(y) \right] - \hat{j} \left[\frac{\delta}{\delta x}(z) - \frac{\delta}{\delta z}(x) \right] + \hat{k} \left[\frac{\delta}{\delta x}(y) - \frac{\delta}{\delta y}(x) \right] \\
&= \vec{A} \cdot \hat{i} [0 - 0] - \hat{j} [0 - 0] + \hat{k} [0 - 0] \\
&= \vec{0} \text{ Answer}
\end{aligned}$$

Q# 68: If $\vec{v} = \vec{\omega} \times \vec{r}$, Prove $\vec{\omega} = \frac{1}{2} \text{curl } \vec{v}$, Where $\vec{\omega}$ is a constant vector.

Answer:

$$\text{Let } \vec{r} = x \hat{i} + y \hat{j} + z \hat{k}, \vec{\omega} = \omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}$$

$$\text{Then } \text{curl } \vec{v} = \vec{\nabla} \times \vec{v}$$

$$\text{curl } \vec{v} = \vec{\nabla} \times (\vec{\omega} \times \vec{r})$$

$$\text{curl } \vec{v} = (\vec{r} \cdot \vec{\nabla}) \vec{\omega} - \vec{r} (\vec{\nabla} \cdot \vec{\omega}) - (\vec{\omega} \cdot \vec{\nabla}) \vec{r} + \vec{\omega} (\vec{\nabla} \cdot \vec{r})$$

$$\text{curl } \vec{v} = (x \hat{i} + y \hat{j} + z \hat{k}) \cdot \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{r} (\vec{\nabla} \cdot \vec{\omega}) - (\vec{\omega} \cdot \vec{\nabla}) \vec{r} + \vec{\omega} (\vec{\nabla} \cdot \vec{r})$$

$$\text{curl } \vec{v} = \left(x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{r} (\vec{\nabla} \cdot \vec{\omega}) - (\vec{\omega} \cdot \vec{\nabla}) \vec{r} + \vec{\omega} (\vec{\nabla} \cdot \vec{r})$$

$$\begin{aligned}
\text{curl } \vec{v} &= \left(x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{r} \left(\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k} \right) \cdot (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) \\
&\quad - (\vec{\omega} \cdot \vec{\nabla}) \vec{r} + \vec{\omega} (\vec{\nabla} \cdot \vec{r})
\end{aligned}$$

$$\text{curl } \vec{v} = \left(x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{r} \left(\frac{\delta}{\delta x} \omega_1 + \frac{\delta}{\delta y} \omega_2 + \frac{\delta}{\delta z} \omega_3 \right)$$

$$- (\vec{\omega} \cdot \vec{\nabla}) \vec{r} + \vec{\omega} (\vec{\nabla} \cdot \vec{r})$$

$$\text{curl } \vec{v} = \left(x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{r} (0 + 0 + 0) - (\vec{\omega} \cdot \vec{\nabla}) \vec{r} + \vec{\omega} (\vec{\nabla} \cdot \vec{r})$$

$$\text{curl } \vec{v} = \left(x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \vec{0} - (\vec{\omega} \cdot \vec{\nabla}) \vec{r} + \vec{\omega} (\vec{\nabla} \cdot \vec{r})$$

$$\text{curl } \vec{v} = \left(x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z} \right) (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - (\vec{\omega} \cdot \vec{\nabla}) \vec{r} + \vec{\omega} (\vec{\nabla} \cdot \vec{r})$$

$$\begin{aligned}
\vec{\text{curl}} \vec{v} &= (x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z})(\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) \cdot (\frac{\delta}{\delta x} \hat{i} + \frac{\delta}{\delta y} \hat{j} + \frac{\delta}{\delta z} \hat{k}) \vec{r} \\
&\quad + \omega(\vec{\nabla} \cdot \vec{r}) \\
\vec{\text{curl}} \vec{v} &= (x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z})(\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - (\omega_1 \frac{\delta}{\delta x} + \omega_2 \frac{\delta}{\delta y} + \omega_3 \frac{\delta}{\delta z}) \vec{r} + \omega(\vec{\nabla} \cdot \vec{r}) \\
\vec{\text{curl}} \vec{v} &= (x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z})(\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - (\omega_1 \frac{\delta}{\delta x} + \omega_2 \frac{\delta}{\delta y} + \omega_3 \frac{\delta}{\delta z})(x \hat{i} + y \hat{j} + z \hat{k}) + \omega(\vec{\nabla} \cdot \vec{r}) \\
\vec{\text{curl}} \vec{v} &= (x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z})(\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - (\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) + \omega 3 \quad [\because (\vec{\nabla} \cdot \vec{r}) = 3] \\
\vec{\text{curl}} \vec{v} &= (x \frac{\delta}{\delta x} + y \frac{\delta}{\delta y} + z \frac{\delta}{\delta z})(\omega_1 \hat{i} + \omega_2 \hat{j} + \omega_3 \hat{k}) - \omega + 3\omega \\
\vec{\text{curl}} \vec{v} &= (x.0 + y.0 + z.0) + 2\omega \\
\vec{\text{curl}} \vec{v} &= 0 + 2\omega \\
\vec{\text{curl}} \vec{v} &= 2\omega \\
\text{Therefore, } \vec{\omega} &= \frac{1}{2} \vec{\text{curl}} \vec{v} \quad (\text{Proved})
\end{aligned}$$

Q# 69: Show that $\phi(x, y, z)$ is any solution of Laplace's equation. Then $\nabla\phi$ is a vector which both solenoidal and irrotational.

Answer:

We have, A solenoidal vector field satisfies $\vec{\nabla} \cdot \vec{B} = 0$

A vector field $\vec{V}$ is said to be *irrotational* if its curl is zero. That is, if $\vec{\nabla} \times \vec{v} = 0$.

A conservative vector field is also **irrotational**.

Since $\phi(x, y, z)$ satisfies the Laplace's equation hence, $\nabla^2\phi = 0$ or $\nabla \cdot \nabla\phi = 0$

Therefore, $\vec{\nabla} \phi$ is solenoidal.

$$\text{and also } \vec{\text{curl}} \vec{v} = \vec{\nabla} \times (\vec{\nabla} \phi) = \left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \times \left(\hat{i} \frac{\delta}{\delta x} + \hat{j} \frac{\delta}{\delta y} + \hat{k} \frac{\delta}{\delta z} \right) \phi$$

$$\begin{aligned}
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ \frac{\delta\phi}{\delta x} & \frac{\delta\phi}{\delta y} & \frac{\delta\phi}{\delta z} \end{vmatrix} \\
&= \hat{i} \left(\frac{\delta^2\phi}{\delta y \delta z} - \frac{\delta^2\phi}{\delta z \delta y} \right) - \hat{j} \left(\frac{\delta^2\phi}{\delta x \delta z} - \frac{\delta^2\phi}{\delta z \delta x} \right) + \hat{k} \left(\frac{\delta^2\phi}{\delta x \delta y} - \frac{\delta^2\phi}{\delta y \delta x} \right)
\end{aligned}$$

$$\begin{aligned}
&= \hat{i} \times 0 - \hat{j} \times 0 + \hat{k} \times 0 \\
&= 0
\end{aligned}$$

Hence $\vec{\nabla} \phi$ is also irrotational. (Proved)

Q# 70: If $\vec{A}$ and $\vec{B}$ are irrotational then prove that $\vec{A} \times \vec{B}$ is solenoidal.

Answer: Since $\vec{A}$ and $\vec{B}$ are irrotational, hence $\vec{\nabla} \times \vec{A} = 0$ and $\vec{\nabla} \times \vec{B} = 0$
and if $\vec{A} \times \vec{B}$ is solenoidal then $\vec{\nabla} \cdot (\vec{A} \times \vec{B}) = 0$

$$\begin{aligned}
\text{L.H.S. } \vec{\nabla} \cdot (\vec{A} \times \vec{B}) &= \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B}) \quad [\because \vec{\nabla} \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\vec{\nabla} \times \vec{A}) - \vec{A} \cdot (\vec{\nabla} \times \vec{B})] \\
&= \vec{B} \cdot 0 - \vec{A} \cdot 0 \\
&= 0 \text{ (Proved)}
\end{aligned}$$

Hence $\vec{A} \times \vec{B}$ is solenoidal. (Proved)

Q# 71: Prove that $(A \times B) \cdot (B \times C) \times (C \times A) = [ABC]^2$

Solution:

L.H.S

$$(A \times B) \cdot (B \times C) \times (C \times A)$$

$$\text{let, } B \times C = X$$

$$\begin{aligned}
\therefore (A \times B) \cdot (B \times C) \times (C \times A) &= (A \times B) \cdot (X) \times (C \times A) \\
&= (A \times B) \cdot [(X \cdot A)C - (X \cdot C)A]
\end{aligned}$$

$$[\text{From Q \# 43, } \vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}]$$

$$\begin{aligned}
\therefore (A \times B) \cdot (B \times C) \times (C \times A) &= (A \times B) \cdot [(B \times C \cdot A)C - (B \times C \cdot C)A] \text{ -----(i)} \\
&[\because B \times C = X]
\end{aligned}$$

Now, $\vec{B} \times \vec{C}$

$$\begin{aligned}
&= (b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}) \times (c_1 \hat{i} + c_2 \hat{j} + c_3 \hat{k}) \\
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \\
&= \hat{i}(b_2 c_3 - b_3 c_2) - \hat{j}(b_1 c_3 - b_3 c_1) + \hat{k}(b_1 c_2 - b_2 c_1)
\end{aligned}$$

$$\therefore B \times C \cdot C$$

$$\begin{aligned}
&= [\hat{i}(b_2c_3 - b_3c_2) - \hat{j}(b_1c_3 - b_3c_1) + \hat{k}(b_1c_2 - b_2c_1)] \cdot (c_1\hat{i} + c_2\hat{j} + c_3\hat{k}) \\
&= c_1(b_2c_3 - b_3c_2) - c_2(b_1c_3 - b_3c_1) + c_3(b_1c_2 - b_2c_1) \quad [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1] \\
&= c_1b_2c_3 - c_1b_3c_2 - c_2b_1c_3 + c_2b_3c_1 + c_3b_1c_2 - c_3b_2c_1 \\
&\therefore \mathbf{B} \times \mathbf{C} \cdot \mathbf{C} = 0 \text{ -----(ii)}
\end{aligned}$$

From (i)

$$\begin{aligned}
\therefore (\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{B} \times \mathbf{C}) \times (\mathbf{C} \times \mathbf{A}) &= (\mathbf{A} \times \mathbf{B}) \cdot [(\mathbf{B} \times \mathbf{C} \cdot \mathbf{A})\mathbf{C} - (\mathbf{B} \times \mathbf{C} \cdot \mathbf{C})\mathbf{A}] \\
&= (\mathbf{A} \times \mathbf{B}) \cdot [(\mathbf{B} \times \mathbf{C} \cdot \mathbf{A})\mathbf{C} - 0] \quad [\mathbf{B} \times \mathbf{C} \cdot \mathbf{C} = 0; \text{ from (ii)}] \\
&= (\mathbf{A} \times \mathbf{B}) \cdot [(\mathbf{B} \times \mathbf{C} \cdot \mathbf{A})\mathbf{C}] \\
&= [\mathbf{A} \times \mathbf{B} \cdot \mathbf{C}][\mathbf{B} \times \mathbf{C} \cdot \mathbf{A}] \\
&= [\mathbf{ABC}][\mathbf{ABC}]
\end{aligned}$$

We have, Scalar triple product: $\vec{A} \cdot (\vec{B} \times \vec{C})$ or $\vec{B} \cdot (\vec{C} \times \vec{A})$ or $\vec{C} \cdot (\vec{A} \times \vec{B})$ are known as a scalar triple product. It is symbolically denoted by $[\mathbf{ABC}]$ or $[\mathbf{BCA}]$ or $[\mathbf{CAB}]$

$$= [\mathbf{ABC}]^2$$

(Proved)